

# Monitoring Technology: The Impact of Body-Worn Cameras on Citizen-Police Interactions\*

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**Abstract:** We provide experimental evidence that using body-worn cameras (BWCs) for police monitoring improves police-citizen interactions. In an intervention carried out in Brazil in 2018, we find that treated incidents show a 61.2% decrease in police use of force and a 47.0% reduction in adverse interactions, including handcuff use and arrests. The use of body-worn cameras also significantly improves the quality of police reporting. The rate of incomplete reports dropped by 5.9%, which is accompanied by a 69.2% increase in reported incidents of domestic violence. We explore various mechanisms that explain *why* BWCs work and show that the results are consistent with the police changing their behavior in the presence of cameras. Overall, results show that the use of body-worn cameras de-escalates conflicts.

**Keywords:** police, use of force, technology, field experiment

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# 1 Introduction

Globally, seven in ten adults trust the police, but in Latin America, a region plagued by high levels of violence and crime, this confidence drops to half (Gallup, 2023). In this setting, body-worn cameras (henceforth, BWCs) are seen as a promising technological solution to increase scrutiny and oversight of the police. By recording encounters that officers have with citizens, these devices create a record of interactions that may bring transparency and accountability to the actions of the police, which could be beneficial where and when violence undermines public trust in the police.<sup>1</sup>

In this study, we provide experimental evidence demonstrating that BWCs used by police significantly enhance the quality of interactions between police officers and citizens. Our study was conducted in the state of Santa Catarina, Brazil, in 2018, and our setting is representative of other countries in Latin America. We randomized police officers to the use of body-worn cameras, inducing experimental variation of body-worn cameras at the *event* level which we view as the “natural” unit of analysis for treatment delivery and outcome measurement, as it is the level at which citizens and police interactions unfold and the use of force or its (de-)escalation may occur.

We investigate the effects of body-worn cameras on different police use of force outcomes following our pre-analysis plan (PAP), and provide further exploratory analysis of officer reporting behavior.<sup>2</sup> Our first set of results pertain to what officers report from the incidents.<sup>3</sup> We can address this because we have access to the internal logs of incidents by police officers, which are not public and are only used as internal registers of police activity. Our findings indicate that

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<sup>1</sup>In our study, we use the term “citizens” to describe any individual who is not actively serving in the police force. This usage is a slight abuse of the term because those interacting with the police may not necessarily be Brazilian citizens.

<sup>2</sup>See Appendix Section C for details about the pre-registration and minor differences between the analysis presented here and the one contained in the PAP.

<sup>3</sup>Throughout the paper, we interchangeably use the terms a police “incident” and “event”, which includes dispatcher-initiated calls, officer-initiated actions, proactive patrols, and incidents without identifiable interactions or victims that have been logged by the police.

incidents involving cameras were 2.77 percentage points less likely to lack essential descriptive information when submitted to police records. Relative to the mean of control, where events without accurate description represent the majority of internal reports, this equates to a decrease of 5.9% in underreporting. In turn, the drop in incomplete reports translate into an increase in the reporting of other crimes, including a substantive increase in reporting of domestic violence. We find that the presence of cameras induce a significant 1.14 percentage point increase in the reporting of domestic violence incidents, which accounts for a 69.2% increase relative to the control mean. This effect is large when compared to interventions aimed at either improving domestic violence reporting (Miller and Segal, 2019; Amaral et al., 2021; Levy and Mattsson, 2023; Colagrossi et al., 2023). We further find that it becomes 9.5% more likely that the incident is referred to investigative bodies, and police reports, on average, are 20.1% more likely to include victims. These effects can be also be a downstream consequence of the increase in the incidence of domestic violence reporting. We expand on this in Section 3.2. Overall, we interpret that officers are more diligent in reporting when in the presence of cameras, especially in cases where video footage can be used as evidence to prosecute. Furthermore, accurate reporting by police officers has been demonstrated to improve the rate at which crimes are cleared (Blanes i Vidal and Kirchmaier, 2018).

Our design introduced random variation both on *whether* a BWC was present on an incident and on *who* carried it. Hence, it allows us to study whether the characteristics of the officer carrying it mattered. We find evidence of increased compliance with the police’s BWC standard operating procedures if the officer wearing the camera is relatively junior. Although we cannot reject the null of equality, we also find suggestive evidence of stronger reporting behavior by junior officers. There are several possible explanations for this finding. It is conceivable that junior officers are more adept at using a new technology. It may also suggest that juniors’ dynamic incentives and career concerns may be important factors driving their effect and that low-rank officers with BWCs monitor their higher-ranked peers.<sup>4</sup> We discuss this and how we interpret the change in police

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<sup>4</sup>In other contexts, peer monitoring in the workplace has been shown to alleviate principal-agent problems. See Bandiera et al. (2009, 2010), Mas and Moretti (2009), and Ashraf and Bandiera

reporting activity vis-à-vis changes in citizen behaviour in Section 3.2.

We then turn our attention to the effects of BWCs on use of force. We find that treated incidents saw a decline in the likelihood of use of force by 61.2%. An “Adverse Interaction Index” following Anderson (2008)—which also combines charges of contempt, disobedience or citizen resistance, and use of handcuffs or arrests— was reduced by 47%.

Consistent with the results on reporting, we find that the reduction of use of force is higher when junior officers are wearing the cameras. In line with the PAP, we then dig deeper on other mechanisms that are specifically relevant to use of force. We find that treatment effects are primarily concentrated in events that, *prior to police being dispatched*, were classified as relatively *low risk* by virtue of there being either no weapons reported on the scene, there being no injuries, or there being any material risk of general unrest as judged by the police. This suggests that cameras affect the situation dynamic by *preventing* the escalation of tension that would counterfactually unfold. We also find that BWCs have larger effects in areas with a higher baseline use of force. Importantly, these latter effects are shown to be statistically different from areas with lower incidence. This suggests that cameras may have the highest benefits in places where police-citizen interactions are relatively strained at the baseline.

Lastly, we document that our estimates are robust to several alternative confounding mechanisms, which were pre-registered. First, we test if officers with BWCs self-select into specific events when wearing cameras. As a virtue of the implementation, dispatch operators were blind to the treatment status of patrol units, limiting the potential for such a selection to happen. Yet, we confirm the absence of endogenous sorting through several empirical tests. More specifically, we show that treatment officers are not more likely to patrol less risky or wealthier areas, that they are not less likely to initiate interactions with individuals (e.g. stop-and-frisk events), and that they do not take longer to respond to the calls. Second, we find similar results in alternative estimation samples. Third, we re-evaluate the effects of BWCs using only observational variation in a differences-in-differences setting and data from precincts that did not participate in the (2018) for a literature review.



intervention. Again, we find similar effect sizes.

This paper is hardly the first one to study BWCs (see [Lum et al., 2020](#), [Williams et al., 2021](#) and [Petersen and Lu, 2023](#) for reviews and meta-analyses, and the recent work of [Ferrazares, 2023](#) and [Kim, 2024](#)). Yet, unlike much of the existing literature, which has almost exclusively focused on the United States or the United Kingdom, we provide robust evidence that body-worn cameras reduce police use of force in Brazil, a large middle-income country that is representative of other localities in Latin America facing high levels of violence. Related to this work, the observational study by [Monteiro et al. \(2022\)](#) finds that BWCs reduced police use of lethal force through the staggered implementation of the devices across police precincts in São Paulo.<sup>5</sup>

We also contribute to the body-worn camera literature by not only showing that BWCs work but also uncovering mechanisms for why it that happens. The combination of granular incident data and a randomized design allows us to explore outcomes and police-side heterogeneities that had not been previously considered. In Section 4 we also reconcile why previous literature has produced mixed results of body-worn cameras in use-of-force. Some past papers found little or no effects of BWCs (see, for example, [Yokum et al., 2019](#)), while other studies showed that devices are effective in curbing the use of force. We shed light on why the literature has found diverging results. Naturally, differences could arise because of the different contexts in which the studies were conducted. Yet, because our granular study nests a broad class of commonly used evaluation strategies, we are able to show that the design and the coarseness of previous studies are the likely culprits.

This paper contributes to a broad literature about police interventions that aim to build trust, improve citizen relations or reduce crime – often through deploying equipment or new information technology to the police. Studying the police, and in line with this study, [Garicano and Heaton \(2010\)](#) and [Mastrobuoni \(2020\)](#) shows that access to IT equipment can improves police perfor-

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<sup>5</sup>[Magaloni et al. \(2023\)](#) who also studied a BWC randomized controlled trial in a neighborhood of Rio de Janeiro, Brazil, but noted very low compliance and minimal camera footage being produced.

mance, especially when complemented by managerial practices. [Bove and Gavrilova \(2017\)](#), [Harris et al. \(2017\)](#) and [Mello \(2019\)](#) show that investment in police equipment, including weapons, has a positive effect on citizen-officer interactions, reducing both complaints and assaults against officers. In criminal justice more broadly, [Di Tella and Schargrodsky \(2013\)](#), [Doleac \(2017\)](#) and [Anker et al. \(2021\)](#) show that criminal (re)offense can be reduced by investments in IT technology such as electronic monitoring and DNA databases.

We contribute to a broader debate on the productivity effects of monitoring actions and oversight of the police. Several studies have shown that the combination of increased investigatory oversight ([Long, 2019](#); [Prendergast, 2001](#); [Devi and Fryer, 2020](#); [Rozema and Schanzenbach, 2023](#)) or through media exposure or scandals ([Shi, 2009](#); [Mastrorocco and Ornaghi, 2024](#); [Prendergast, 2021](#); [Cheng and Long, 2022](#); [Ang et al., 2024](#)) can generate substantial responses from police officers who may reduce effort and partially withdraw from law-enforcement activities. This paper shows that a monitoring technology can indeed improve police performance without necessarily inducing police passivity or depolicing. In this sense, the results in this paper align more closely with [Rivera and Ba \(2023\)](#) who show that improved oversight without public exposure can improve behaviour without crime effects.

Finally, a number of papers highlighted how public and police officer personnel productivity is affected by various reasons beyond oversight, such as pay considerations ([Mas, 2006](#)), workplace incentives ([Cheng and Long, 2018](#); [Ornaghi, 2019](#); [Chalfin and Gonçalves, 2023](#)), ethnicity ([Vanden Eynde et al., 2018](#)), and risk perceptions ([Cho et al., 2023](#)), communications ([Battiston et al., 2021](#)), and quality of management ([Banerjee et al., 2021](#); [Fenizia, 2022](#)). More broadly, career incentives in the public sector are also studied by [Xu \(2018\)](#) and [Bertrand et al. \(2020\)](#).

We proceed as follows. Section 2 provides the context, presents details about the intervention, and discusses the data and measurement approach. Section 3 provides the main results and robustness checks that could threaten the validity of our estimates. We also include a welfare analysis and show that the Marginal Value of Public Funds (MVPF) compares favourably to other police reform interventions. Section 4 situates our results in the light of previous literature and corroboration.

rates our experimental evidence using observational data. Section 5 concludes and discusses policy implications of the experiment results.

## 2 Context, Intervention, and Data

**Context.** Brazil is one of the most violent countries in the world. In 2019, the homicide rate was 20.8 homicides per 100 thousand inhabitants compared to 5.0 and 1.2 in the US and the UK, respectively.<sup>6</sup> We implemented the BWC intervention in the state of Santa Catarina, one of Brazil’s least violent states. Yet, it is still only comparable to the most violent US states. Appendix Figure A1 shows homicide rate per 100,000 inhabitants for Brazilian and American states, and other major Latin American countries for a comparison. Santa Catarina (SC) exhibits a homicide rate two times higher than the US, and is broadly comparable to Latin American countries, where levels of violence are higher.<sup>7</sup> We collaborated with the Igarapé Institute and the Santa Catarina state Military Police (PMSC), the main police body responsible for patrolling, responding to emergencies, and operating the 911 hotline. It is the most visible element of the policing institutional infrastructure in Brazil. Our experiment started in September 2018, lasted three months, and included five police precincts: Florianópolis, São José, Biguaçu, Tubarão and Jaraguá do Sul. Those sites were chosen given their easy accessibility from the police headquarters in Florianópolis and to represent a variety of settings in terms of socio-demographic characteristics and baseline violence levels.<sup>8</sup>

**Intervention.** In our context, police body-worn cameras were implemented under a Standard Operating Protocol requiring that cameras be activated during all interactions with citizens (Polícia

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<sup>6</sup>See United Nations Crime Trends Survey, available at <https://dataunodc.un.org/>.

<sup>7</sup>Appendix Figure A2 compares Brazilian and American states in terms of police killing rates, and shows that overall Brazil is a much more violent context than the US.

<sup>8</sup>A map of the experimental locations is provided in Appendix Figure A3, while Appendix Table A1 studies site demographics. The 5 selected precincts cover for approximately 15% of the state population. In Section 4, we also show that study sites did not present diverging pre-trends from non-experimental precincts.

Militar de Santa Catarina, 2018). While the protocol in principle removes any discretion from the officers, we observe that, in practice, officers retain some discretion in deciding whether to activate the cameras (see also results in Section 3.1). In the context of the United States, previous research has also noted that cameras were implemented under different activation policies (Lum et al., 2020, p.18) and, yet, most agencies allowed for some level of officer discretion, similar to what is observed in the current study.

Body-worn camera interventions can be seen as a shock in the police’s ability to monitor both their own behavior and that of citizens.<sup>9</sup> Several interrelated mechanisms may explain the mechanisms for why they work. First, cameras can alter police officers’ behavior by increasing the perceived likelihood of punishment for deviations from established protocols (Becker, 1976). It has also been suggested that body-worn cameras may raise officer self-awareness, which can moderate their actions and behaviors (Ariel et al., 2018). Second, cameras may also prompt behavioral responses from citizens who, aware that the interaction is being recorded, may be less likely to engage in confrontations and more inclined to de-escalate, thereby reducing the need for police use of force from the outset.

We note that while body-worn cameras increase the ability of the police to monitor *itself*, it does not necessarily increase *external* investigatory oversight. This differs from previous studies which used variation from monitoring of the police by outside bodies or through media exposures, and generally concluded that increased oversight reduced police effort.<sup>10</sup> Although unquestionably some recordings do reach the public, they represent only a small minority of cases due to strict controls and footage access.<sup>11</sup> Body-worn cameras interventions further differs by increasing *ex-ante expectation* of monitoring, as opposed to *ex-post* oversight that may ensue from those ex-

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<sup>9</sup>We also note that police officers typically supervised the introduction and utilization of the devices. Videos were proactively reviewed, used in operational assessments, served as training materials, and officers had a reasonable expectation of such.

<sup>10</sup>See for example Prendergast (2021), Cheng and Long (2022), Ang et al. (2024), and others.

<sup>11</sup>See <https://www.nytimes.com/2023/12/13/magazine/police-body-cameras-miguel-richards.html> for examples where videos generated by the cameras reached media attention.

traordinary events. We then argue that body-worn cameras are likely to trigger different responses without public reaction in general, and in this sense aligns more with the effects reported by [Rivera and Ba \(2023\)](#).

**Design.** The ideal experiment to evaluate the effects of BWCs would randomize the presence of cameras to incidents. This is logistically and operationally infeasible in the context of policing. Alternatively, we designed an experiment that seeds random allocation at the incident level.

Figure 1 provides an overview of the experimental design, starting with the project timeline in Panel A. The randomization of officers was drawn about a month before the experimental period, and the equipment installation and training took place the week before the experiment started, and the intervention lasted for three months. Panel B illustrates how we inserted the into the IT infrastructure that was only possible due to our unique depth of integration with the police. In this way, the randomization between officers induced variation at the incident level. The table illustrates the incident-level data including in the pre-intervention period. Hypothetical officers {A, B, C and D} combine in pairs to attend an incident (along rows).<sup>12</sup> We consider an incident as treated if at least one of the police officers was wearing a camera. Thus, less than half of police officers have to be treated in order to achieve a 50% treatment at the *incident level*. For example, in the hypothetical example, if only officer A is treated, that achieves the desired treatment allocation at the incident level. If two officers are treated (out of four), almost no variation may be left at the incident level.

[Figure 1 here]

In practice, we simulated how many officers would have to be treated in order to have around 50% of the events treated using real pre-intervention incident data. Panel B in Figure 1 shows the simulation result. They indicate that between one-in-four and one-in-three treated officers would achieve half of treated incidents (dashed horizontal reference line).

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<sup>12</sup>In reality, the modal incident groups size is indeed two, but there is variation in the group size.

In the end, we preferred to adopt a one-in-three design (dashed vertical reference line). Out of the roster of sworn police officers per precinct we obtained in July 2018, we randomly selected 1/3 of the officers to be in the treatment group and 2/3 to the control group. We have 150 officers assigned to wear BWCs and 295 control group officers. We stratified by precinct, officer activity, rank, previous internal investigations, and gender (40 stratification bins).

Besides randomizing which officers wear a BWC or not, we introduce a second layer of randomization. Every week during the fourteen weeks of the experiment, two days were randomly selected to serve as blackout days, with the randomization stratified by day of the week, providing us with across-shift variation to camera exposure. Treated officers always wear a camera if their 12-hour shift falls on days that – due to our second layer of randomization – were not selected to serve as *blackout days*. We leverage those to assess the persistence of camera effects and have across shift variation (see Section 4). Control officers were mandated not to wear a camera on any shift.

Thus, the design induces random allocation of cameras at police incidents, our primary unit of analysis. We consider our treatment to be exposure to cameras at the event level – that is if there is at least one officer in it wearing a camera. Since the vast majority of incidents involve more than one officer, our sparser one-in-three officer-level randomization was calibrated such that approximately half of the incidents post-treatment would have a body-worn camera, maximizing power and in sharp contrast with the existing literature, which typically assigns cameras to more than 50% of the officers participating – we will elaborate on this in our discussion about the literature in Section 4.

Figure A4 displays the number of incidents by day over the project period along with a moving average of both the total number of incidents and the number of treated incidents, showing that we successfully induced around half of the events treated. Out of the population of events that did not occur on blackout days, around 58% have had an officer present wearing a BWC, in line with our simulations.

**Integrity of the research design.** The integrity of the research design was protected by a host of precautions. Cameras and docking stations were kept in the armory of the police precincts that officers visit at the start and end of each shift to collect and return their service weapons and equipment. Further, the blackout days were randomly selected at the beginning of the experiment but only communicated directly to the armory the evening before to avoid potential selection around the blackout days. Moreover, dispatch operators were blind to whether units were carrying a BWC. We find no evidence suggesting significant non-compliance or other issues that could affect the integrity of the experiment, which we discuss in the robustness checks. We further describe the implementation details in Appendix A.

Throughout the implementation, the research team had strong backing from the police leadership. Following recommendations on best practice that were informed by past research suggesting low compliance with BWC use (which we discuss in Section 4 in more detail), a standard operating protocol (SOP) was developed mandating that every incident involving an interaction with a citizen should be recorded, with few exceptions, such as sensitive or covert operations. Officers were required to inform citizens verbally that “the incident was being recorded, according to police protocol,” whenever the situation allowed.

The research team never had access to recordings due to individuals’ privacy concerns and the sensitive nature of such data. We were nevertheless able to measure compliance at the individual incident level, as we outline further below.

**Data and outcome measurement** We primarily draw on *incident-level data*, which was facilitated by PMSC’s fully digital backend called PMSC mobile. The data captures the universe of all events that were attended by any PMSC officers. One important aspect of the incident-level data is that PMSC mobile is a state-wide technology and the definitions of variables are always the same, both across space and over time. The system was already in place before our experiment, so officers did not have to learn how to use a new system together with the use of cameras. The dataset is kept for internal use by the police, and they have granted us access for this study.

Events in the PMSC mobile system typically originate from self-initiated calls due to routine

operations (such as patrolling) or due to scheduled activities (e.g., the execution of court orders), although 91.8% of them are the result of police being dispatched to a 911 call of service through the central dispatch service. Our main outcome dataset contains a total of 17,665 events that span the experimental period ranging from September 3, 2018 to December 10, 2018.

The incident data includes a number of useful fields. It reports the time of the incident (arrival at the scene and end of the event), hashed officer identifiers (allowing us to link to the treatment status) and precise geo-coded locations and addresses. It further contains the information about the serial number of the cameras that officers were wearing during the incident, if any. Those are merged with the individual camera log files, which provide both the serial number of the device and all information on when and for how long the camera was activated. From this process, we created our measure of whether recordings actually took place.

At the end of the incident, officers fill in a report with their observations and descriptions about what unfolded in the operation. We use this information as part of our exploratory analysis on how increased monitoring induced by the cameras affect officer reporting behavior, with results presented in Section 3.2. Our first measure of reporting is whether the police officers filled in a complete report.<sup>13</sup> Conditional on observing information about the nature of the events, these reports describe the crime type (out of those that have any crime type reported, the most common are verbal attrition or threat, noise complaints, burglary, assault, and domestic violence), an indicator for the presence of victims, and a field that indicates whether the incident was later handed over to the investigative unit (“Policia Civil”). We also made use of these reporting measures.

Following our pre-registration, we also observed the types of force that were deployed in the incident (physical, non-lethal, or lethal force). Note that use of force is self-reported by the police

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<sup>13</sup>In fact, this is important, as the events frequently had key details empty and/or incomplete (47.3%).



officers through the police systems.<sup>14,15</sup> We observe if handcuffs or arrests were deployed. Our final measure of use of force pertains to contempt, disobedience or resistance charges towards police officers. We further created an inverse covariance-weighted index combining these three outcomes following [Anderson \(2008\)](#), which we call Adverse Interaction Index.<sup>16</sup>

Important to some heterogeneous analyses that follow, the central dispatch service pre-classifies the risk of the event. This assessment is done prior to officers being dispatched to the event, so it is not contaminated by the presence of cameras. An event was classified as high risk if any of the following conditions are met: there are individuals with life-threatening injuries, the suspect is still on site, the suspect was armed, and there is a general risk of broader disturbance to peace. An event was considered low risk if the response is negative to all these questions.

In addition to the incident data, we further observe a range of officer characteristics, such as their job title, rank, gender, the date of admission to the police, and the number of internal investigations that have involved the specific officer. These characteristics were also used to inform the stratification of the camera random assignment and were further used to explore pre-registered heterogeneous effects.

### 3 Empirical Framework and Results

In this section, we present the empirical framework and the main results and discuss range of further empirical tests to speak to the robustness of the results to guide our interpretation of the main mechanisms driving the effects. In what follows, we use the following empirical specification

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<sup>14</sup>As we discuss in Section 3, it is conceivable that the presence of the cameras force officers to report the facts more accurately. In this case, we would expect that the presence of cameras reduce the under-reporting of use of force, and would thus bias the estimates *against* finding any effect.

<sup>15</sup>The reports disclose if use of force is deployed, but they do not contain information about which officers have used force.

<sup>16</sup>The use of force outcomes were registered in the pre-analysis plan. The reporting margin is considered part of an exploratory analysis. We detail the pre-registration of this study in Appendix Section C.

to study the effect of the presence of BWCs across a set of outcomes measured at the police incident level:

$$y_{ibdw} = \beta \times \text{Treated Event}_i + \eta_{bw} + \tau_d + z_{ibdw} + \sum_{j=1}^n \phi_{o_j(i)} + \epsilon_{ibdw} \quad (1)$$

In this specification,  $i$  indicates an event attended by the police,  $b$  is the police precinct,  $d$  is the day of the week, and  $w$  is the week of intervention. For our main specifications,  $\text{Treated Event}_i = 1$  if at least one officer forming the unit that attended event  $i$  was assigned to wear a camera. This implies that our estimates capture an intention-to-treat effect. We include police precinct-by-week fixed effects ( $\eta_{bw}$ ), day-of-the-week fixed effects ( $\tau_d$ ), and officer  $o$  stratification bins ( $\sum_{j=1}^n \phi_{o_j(i)}$ ). We also control for the number of officers involved in the event  $z_{ibdw}$ . In our initial specifications, we exclude blackout days and focus exclusively on comparing treated with control events.<sup>17</sup> The disturbance  $\epsilon_{ibdw}$  is clustered at the police precinct-by-week level. We also include randomization inference p-values that are free from assumptions about the structure in the disturbance term.

### 3.1 Recording

Column (1) of Table 1 shows the frequency in which the cameras were activated by police officers in the incidents they attended. We show that, on average, 24% of the treated events were recorded. Virtually none of the control group events had any camera recording linked to them. To interpret those coefficients, note that the SOP required the use of cameras only if there were interactions with citizens, which does not occur in all incidents. To keep some perspective, 47.3% of incidents in the control group were reported without any information regarding the nature of the infraction. Assuming that there was no interaction with citizens in those incidents, only the 52.7% of events could be recorded if every incident was treated.

[Table 1 here]

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<sup>17</sup>Our main sample excludes shifts without cameras to correctly estimate the average treatment effect of the BWC. We later include the blackout days in the sample to study if officers change their behavior even in the absence of a camera.

**Heterogeneity by officer rank** Panel B of Table 1 reflects whether the effect is heterogeneous by officer rank. We seek to understand how supervision and managerial relations are key to understanding workers’ performance in our context (Bandiera et al., 2009; Frederiksen et al., 2020; Fenizia, 2022). We leverage the fact that not only the event-level exposure to the camera was random but also the officer who was carrying it, allowing us to explore heterogeneous effects by their characteristics. Given the military hierarchy structure at PMSC, we classify police officers in two categories, either a low-ranked “soldier” category or a higher-ranked category for corporal or above ranks. Such rank distribution, however, is not observable to the average civilian. Importantly, compliance with the protocol was lower when higher-ranked officers carried the camera: 25% of incidents were recorded when junior officers ported the camera, as compared to 18.9% when senior officers wore the device, a difference of 6.1 percentage points or 32.3%.

There are several possible explanations that can rationalize this effect. First, it’s conceivable that junior officers are more adept at using technology and adjust more swiftly to new technological tools. Second, this observation aligns with the officers’ roles, where junior officers tend to be more actively engaged with citizens, whereas more experienced officers, though at a similar physical distance, often take on more of a supervisory roles. Third, these effects are in line with a career-concerns motive: early-career officers may be more likely to show behavioral improvements and protocol compliance when in the presence of a camera. Additionally, this insight contributes to existing research by suggesting that the practice of reverse monitoring (junior officers monitoring their senior counterparts) might also play a role in alleviating principal-agency in the delivery of public services.

In common, the three main mechanisms support the idea that police behavior is likely to change when wearing cameras. This does not rule out that citizens also change their conduct when in presence of a camera, but it suggests that it is not *exclusively* citizens changing their conduct. We will return to this point when interpreting the effects on the reporting margin (Section 3.2) and use of force (Section 3.3).

**Heterogeneity by treatment intensity** Panel C documents that the treatment effects are larger with more on-site cameras. This suggests that the extensive and intensive margins of monitoring matter. We find that, relative to incidents with one camera, incidents with two or more cameras were recorded 8 percentage points more often, suggesting, as we will see below, that more intense use of cameras produces higher effects, although with decreasing returns to scale.

### 3.2 Reporting Effects

Police officers complete a report sheet after each incident. They describe the incident's details, including its nature, any potential victims, and information about possible perpetrators. They also collect initial evidence at the scene. This information is then stored in police systems and can be accessed later for investigations and possible legal action. Essentially, this data serves as the starting point for criminal prosecution (Blanes i Vidal and Kirchmaier, 2018). Columns (2)–(9) of Table 1 examine how the use of BWCs influenced what police officers report about the incidents.

Column (2) of Table 1 shows a record of how frequently events were reported with missing the information about their nature. This can naturally occur when, for instance, the police team cannot identify the source of the event, locate individuals, or find evidence related to the reported crime. In practice, instead of assuming that information is purposely missing, this might also indicate that officers are making a conscientious effort to adhere to the established reporting standards of the police department. This accounted for 47.3% of control incidents. The estimates reveal that the presence of BWCs reduced the occurrence of events being reported without data collection by 2.77 percentage points, which is equivalent to a 5.9% decrease, amounting to 368 incidents during the treatment period.

In Columns (3)–(7) deepen the analysis and investigate which crime types were reported more often in treated events. We observe the effects across the five more recurring crime types, of which we report the mean of control events in parentheses: verbal attrition or threat (9.9% of incidents), noise complaints (8.6%), burglary (4.8%), assault (3.6%), and domestic violence (1.6%). According to the randomization inference p-values, we find no significant effects across the first three outcomes, and a marginally significant effect at 10% on assault (with a p-value of 9.6%). In contrast,

we find an increase in the reporting of domestic violence by 1.14 percentage points, highly significant at 1% significance level. Compared to the control group (domestic violence being reported for 1.6% of events), this represents a shift of 69.2% more cases of domestic violence being reported, or 151 cases. This effect is large compared to interventions specifically designed to increase domestic violence reporting, such as increasing female police officer participation (Miller and Segal, 2019), establishing women's police stations (Amaral et al., 2021), or responses to external events like the Me Too movement (Levy and Mattsson, 2023) and news coverage on femicide (Colagrossi et al., 2023).<sup>18</sup> Relatedly, Sviatschi and Trako (2024) finds that the presence of women's justice courts providing police, medical and legal services to victims decreases self-reported domestic violence but doesn't provide any information on police reporting of domestic violence.

The results also suggests that  $\frac{1.14}{2.77} = 41.2\%$  of the reduction in under-reporting is translated into domestic violence cases. We interpret these findings as further evidence that BWCs improve reporting by police officers. This is particularly relevant in cases such as domestic violence, in which, without hard evidence, the prosecution is considerably more challenging.<sup>19</sup>

Column (8) examines the impact of cameras on the share of incidents with reported victims. We discovered that, on average, about 13.8 percent of reports include victims, and the presence of cameras increased this number by 2.8 percentage points, which corresponds to a 20.1% increase. Moving to Column (9), we investigate the effects of police officers wearing BWCs on whether the incident they respond to results in the creation of an investigative report by the investigative police

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<sup>18</sup>For a benchmark, Miller and Segal (2019) suggest that every percentage point of increased female participation in the police force translates into one percent increase in domestic violence reporting. From the baseline of approximately 27% female police officers, extrapolating our effect sizes to their context would imply that achieving a similar impact would require a nearly all-female police force.

<sup>19</sup>Insufficient evidence is identified as one of the underlying causes of the high attrition rates on gender-based violence in the criminal justice process (UNODC, 2014, pp. 38). Graef et al. (2023) also demonstrate the importance of witness appearance in court for case outcomes.

unit ("Polícia Civil"). We observed a significant 9.5 percentage point increase in the likelihood of an investigative report being generated as a result of using BWCs during police operations.

Now, we seek to offer suggestive evidence about how the observed effects on the number of victims and the generation of investigative reports might be linked to a substantial increase in domestic violence cases. It is conceivable that the more frequent reporting of these cases could naturally result in an increase in the number of reported victims and a higher likelihood of investigation. This aspect is crucial in the broader conversation about the overall welfare implications of the reporting impacts of BWCs. As mentioned in the introduction, such increases in reporting are not necessarily welfare enhancing. For instance, as demonstrated by [Monteiro et al. \(2022\)](#), by reducing officers' discretion, the use of BWCs may also lead to more reporting of less significant crimes, like small drug possessions, which could potentially divert police resources from addressing more socially significant crimes and/or increase the prosecution of misdemeanors, leading to demonstrated negative consequences for the lives of those who are prosecuted ([Agan et al., 2023](#)).

To shed light on this issue, we observe that a large share of domestic violence cases reported a victim (73.4%) and were brought to further investigations (75.3%) at the baseline. Thus, it is conceivable that the effects in Columns (8) and (9) of Table 2 are indirect or downstream consequences of the increase in the reporting of domestic violence. To quantify the extent to which this is the case, we employed the following back-of-the-envelope calculation. As reported above, the rescaled treatment effects suggest that an additional 151 domestic violence cases were reported. Under the baseline parameters, this translates to an additional 110.8 cases with victims.<sup>20</sup> In turn, the point coefficients in Column (8) indicate that an additional 369.4 cases reported a victim. Thus, roughly  $\frac{110.9}{369.4} = 30.0\%$  of the main effects could be attributed to a downstream consequence of the increase in domestic violence reporting alone. A similar calculation suggests that 27.4% of the coefficient in Column (9) regarding investigative reports could be attributed to the increase in domestic violence. Overall, this suggests that a sizeable portion of the amplification of the demands

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<sup>20</sup>This is under the assumption that those shares are unchanged in the period after the intervention, which is likely a lower bound considering that cameras improved reporting.

for investigative services were driven by these cases.

**Heterogeneous treatment effects.** The heterogeneous effects in Table 1 suggest that the rank of the officer holding the camera can mediate reporting behaviour, akin to Section 3.1 when discussing the drivers of recording. The results on Panel B indicate that the BWC treatment effects in the reporting margins were only present when an officer with a *soldier* rank was holding the camera in the incident. Treatment effects were null when senior officers were the only ones able to activate the camera.<sup>21</sup> However, we cannot reject the null hypothesis that the effects are different by officer rank. Events treated with higher-ranked officers represent a much smaller share of the total of events (see Appendix Table A2) and we may be underpowered for this analysis.<sup>22</sup> Overall, we take this as suggestive evidence of lower compliance effects and office rank being an important mediator of the camera effects. Panel C suggests that a higher number of cameras on the site was associated with better effects on reporting, although with likely decreasing returns to scale.

### 3.3 Effects on Use of Force

Table 2 displays the primary findings regarding police use of force along different measures. Column (1) shows the impact of the police deploying physical, non-lethal, or lethal force. We note a significant decrease in the use of force in the treated events; 0.69% of the control events reported use of force in comparison to 0.26% of the treated events, corresponding to a reduction of 0.43 percentage points or 61.2%.

[Table 2 here]

Although self-reported use of force data can be subject to underreporting, the presence of cameras is likely to mitigate this issue, as suggested by the improved reporting mentioned earlier. In this case, the reduction of underreporting would be a bias *against* finding treatment effects. That

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<sup>21</sup>The coefficients on reporting of verbal attrition and threat is even negative in those cases.

<sup>22</sup>Although we have approximately one-third of senior officer in our experiments, the events where they are the ones carrying the camera correspond to a total of 1186, or 8.9% of our total event sample as we report in Tables A2 and A3.

is, if police officers were under-reporting their use of force and the BWCs work as an incentive for them to report truthfully, we would interpret the results of Panel A as a lower bound to the true effect of BWCs on use of force.<sup>23</sup> The substantive decline in use of force marks a notable contrast with the existing literature which has mostly found mixed effects. We revisit this divergence from previous work in Section 4.

Columns (2) and (3) substantiates the effects along other margins of use of force between the police officers and citizens. The impact of BWCs on handcuffs or arrests and charges of contempt, resistance, or disobedience. The main effects find negative point coefficients (of 5.9% and 28.2% respectively), although not statistically significant. We then combine all three indicators on the “Adverse Interaction Index,” which revealed a significant and relevant causal estimate of 47.0% of reduction of adverse interactions between citizens and the police.

We interpret these effects as ensuing from the accountability and diligence of the police actions and on-the-scene dynamics promoted by the camera. We next delve into the mechanisms for why this effect takes place, as evidenced by the heterogeneous treatment effects.

**Heterogeneity by officer rank and treatment intensity.** We first reproduce the same heterogeneity as in Panel B of Table 1 in order to observe how the camera effects on use of force are mediated by the officer rank structure.

When interpreting this variation, it is important to note that the salience of the camera to the citizen remains consistent regardless of which officer is wearing it. Both junior and senior officers have close interactions with the public. Therefore, since the camera’s salience to the public remains constant, any differing effects can be attributed to and understood as likely outcomes of changes in police behavior.

In line with Section 3.1, we only find significant effects on all use of force measures when junior officers were required to wear the cameras, with point coefficients generally smaller in absolute value and non-significant for all outcomes. Echoing the compliance and reporting effects, this

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<sup>23</sup>See also [Monteiro et al. \(2022\)](#) for an observational study for São Paulo which estimates comparable effects on the use of force using independent measures of police violence.



confirms that police officer behavior was a likely mediator of the changes in use of force (without eliminating the possibility that there was a behavioral change by the citizens, too). In other words, these results indicate that citizen pacifism cannot be the *only* explanation for why BWCs produce these strong reductions in use of force. We interpret this channel as suggestive evidence, since we cannot formally reject the null of equality in coefficients due to the power issues discussed in Section 3.2.

Panel C documents that the treatment effects were larger with more on-site cameras. Relative to incidents with one camera, incidents with two or more cameras had a reduction in the Adverse Interaction Index – promoting a further drop of 25.9%. In particular, the use of force fell by 79.8% when two cameras were present, also representing how increasing the treatment intensity increased the magnitudes of the effects, although with decreasing marginal returns to scale. This effect, however, should again be interpreted with caution as the *p*-value for the equality of the treatment by its intensity is 0.386 in column (4).

**Heterogeneity by ex-ante risk assessment.** We now extend our analysis beyond the previous tables’ heterogeneity margins to focus on aspects relevant to use of force cases. Panel D studies whether effects were primarily concentrated in events classified *ex-ante* as low risk.

The results presented in Panel D of Table 2 suggest that the effects of BWCs were driven by events that were ex-ante classified as low risk. For those events, the adverse interactions index was reduced by 51%. Columns (1)–(4) show that the point coefficient is negative for all the index components. Only handcuffs or arrests are not statistically significant at the 10% level. No BWC effects are detected among events that were considered to be high-risk ex-ante: the index points to a much smaller and non statistically significant reduction of 8.8%. Only the point coefficient on the use of force is negative, and it is non-statistically significant with large standard errors. Even though we cannot reject the null of equality between the coefficients for the ex-ante risk assessment heterogeneity, we interpret the finding of muted (for the case of the adverse interaction index) and positive (for handcuffs/arrests, and citizen behavior) effects as suggestion that low-risk events are

likely driving our findings in Panel A.<sup>24</sup> Overall, those results indicate that cameras indeed serve as a way to de-escalate conflicts, diffuse tensions, and ensure a better cooperative environment on both sides.

**Baseline use of force.** We explore the extent to which the camera effects are higher in areas with higher likelihood of use of force at the baseline. To do so, for each census tract, we counted the events with use of force in the 13 weeks before the experiment and split the areas along the median for each municipality.<sup>25</sup> The results, depicted in Panel E of Table 2, interact the measure of baseline use of force with the treatment indicator and suggest that stronger treatment effects are observed in areas of the municipalities that experienced higher use of force at the baseline, despite compliance being unchanged across the two areas. The Adverse Interaction Index suggests a reduction in use of force that is nearly five times larger in absolute terms in areas with a historically higher propensity to involve use of force compared to areas with a lower historic use of force. Importantly, we reject the null of equality between these estimates. This is consistent with cameras being effective deterrent devices, especially in places and situations where use of force would be more likely to unfold counterfactually. This indicates that BWCs may be particularly suitable to benefit citizen-police dynamics in areas that historically involved a higher degree of use of force. We consider this to be particularly important when considering how the roll-out of such technology may need to be prioritized in areas with relatively strained citizen-police interactions.

**Blackout specifications.** We next discuss whether BWCs affect the behavior of police officers *after* they had worn the devices in previous days. Our analyses that follow could be suggestive of learning effects taking place. To examine this, we replicate Equation (1) for the blackout days. That is, we compare events that would be treated against those that would be control if counterfactually

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<sup>24</sup>Again, we might be underpowered to reject the null of equality even if those effects are different: high risk events comprise 11.9% of our total sample.

<sup>25</sup>The results are robust to using alternative measures of baseline use of force or geographies.

not on blackout days.<sup>26</sup>

The results are presented in Appendix Tables A4 and A5 for, respectively, effects on reporting and on use of force. Column (1) of Appendix Table A4 highlights that, on blackout days, hardly any event were recorded due to the experimentally induced absence of cameras. We still find that a minority of events were recorded (3.3%), which may have occurred because some officers started their shifts on regular days, and ended during blackout periods. According to experimental protocol, in those cases they would continue to use the cameras due to operational constraints of returning the devices while attending incidents. Despite the much lower recordings, we still find significant effects on decreasing the frequency of reports with no information, and continue to observe increased reporting of verbal attrition and threats. Again mimicking the main results, these events generated investigative effects (although on the margin of significance).

Table A5 demonstrates that all BWC treatment effects on the outcomes measuring citizen and police use of force disappeared. Yet changes in the reporting behavior persisted. This could indicate that officers who were previously exposed to the use of cameras behaved differently, even in the absence of the camera. The learning effects primarily affected the reporting margin but did not appear to have an effect on use of force or other citizen-police interaction margins.

### 3.4 Robustness

**No endogenous allocation of incidents** The observed decrease in the interaction margins could be confounded by a change in the pattern of policing rather than cameras inducing a change in officers' behavior. This could occur, for example, if the treated officers chose to patrol safer areas or areas with less potential for the use of force compared to the control incident units. We show that this hypothesis finds little support in our data in multiple ways.

We test for the absence of endogenous sorting as a function of treatment in an econometric setting. We estimate Equation (1) with characteristics of the event as outcomes and evaluate if the treatment affected them. The results are presented in Table 3. In column (1), we test if officers with

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<sup>26</sup>This is preferable to exploring the variation across days: throughout our sample period, we had only 24 blackout days.

cameras avoided geographic locations with a higher baseline use of force.<sup>27</sup> The results show that cameras were not allocated to events as a function of the baseline level of use of force. Overall, we find no reasons to believe that cameras hinder officers from working in areas where citizen-police interactions are more likely to escalate.

[Table 3 here]

To test the allocation of incidents in space, we regress treated events against latitude and longitude measures in Columns (2) and (3). We find that treated and control events were statistically similar concerning latitude. The point coefficient for longitude is 0.002 of a degree. This distance is negligible, representing roughly 200m measured at the equator; further, while the point estimate appears significant using conventional inference methods, it is insignificant using a randomization inference (p-value of 0.374).

Column (4) captures if the event occurred in a census tract above median levels of income at baseline. A positive coefficient would indicate that officers were more likely to tend to incidents in relatively wealthy neighborhoods when wearing cameras. The effect is statistically insignificant with a randomization inference p-value of 54.1%.

Columns (5) and (6) reflect whether the treatment affected measures such as time to dispatch and an indicator if the time to dispatch was greater than five minutes. The interval between an incident being reported and the officer arriving at the scene was the same between treated and control events. Therefore, in summary, treated and control events occurred in the same places and had the same baseline level of use of force, and treated officers did not take longer to get there.

A potential concern could be that cameras may change the way incidents occur. Most incidents were initiated by the central dispatch and call handlers, who were blind to the underlying treatment status. Nevertheless, there is a potential concern that officers carrying a camera may not initiate events at the same rate as officers without cameras: the regression outcome in Column (7) is a

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<sup>27</sup>This follows the division of the sample between above and below within-precinct median baseline use of force, the same as used in Table 2 Panel E.

dummy variable equal to one if the police self-initiated the incident. We also test whether treatment status is uncorrelated with call-handler-induced incidents: Column (8) confirms this. Columns (7) and (8) together confirm there were no compositional effects in the method through which incidents were initiated. Finally, Column (9) shows that there was no differential assignment of officers with cameras to events based on their ex-ante risk level as measured in Panel D of Table 2.

A potential concern could be that cameras may change the way incidents occur. Most incidents were initiated by the central dispatch and call handlers, who were blind to the underlying treatment status. Nevertheless, there is a potential concern that officers carrying a camera may not proactively initiate events at the same rate as officers without cameras, an effect known in the literature as de-policing and that is particularly pertinent in situations where officers have high discretionary power, such as stop-and-frisk events (Premkumar, 2019): the regression outcome in Column (7) is a dummy variable equal to one if the police self-initiated the incidents, for which we find null effects. We also test whether treatment status is uncorrelated with call-handler-induced incident: Column (8) confirms this. Columns (7) and (8) together confirm there were no de-policing effects from the additional internal monitoring and that incidents were initiated in the same way as control ones. Finally, Column (9) shows that there was no differential assignment of officers with cameras to events based on their ex-ante risk level as measured in Panel D of Table 2.

Appendix Table A6 goes further and explores if the null effects could be heterogeneous by officer rank. It is possible that more experienced officers are more able to anticipate the monitoring effects of the camera and react more strongly by altering their policing patterns. We find that, except for latitude and longitude, the heterogeneous effects are non-statistically significant using randomization inference p-values. The effects on latitude are only significant when the incidents composition had both high- and low-ranked officers but not significant with only high-rank types. Effects on longitude are .009 and .012 degrees with one and two high-ranked officers respectively, significant with randomization inference p-values of 7.0% and 2.9%. The point coefficients are interpretable as a shift of 900m and 1.2km to the east. As the other columns suggest, this spatial change is not correlated with shifting patterns toward places with less baseline use of force, higher

income levels, higher baseline ex-ante risk, a reduction in time to dispatch, or other compositional effects. This difference is not visible in Appendix Figure A5 which shows the incident heatmap by treatment status. In summary, the results suggest that there was no substantial selection in space as a function of the treatment and that officers' behavior with respect to patrolling, as arriving at the event and the working location did not seem to be altered as a function of the cameras.

**No endogenous allocation of senior officers to riskier events** One potential concern is if high-rank officers are more likely to attend higher-risk events, which could have a lower potential for the de-escalation of conflicts. The heterogeneous treatment effects documented in Panel B of Table 2 show that treatment effects were driven by events in which a lower-ranked officer was carrying a BWC. Therefore, this could raise a concern about senior officers being allocated to attend to higher-risk events, thereby confounding the results. Again, our experimental protocol ruled this out as dispatch operators were blind to the respective treatment status of any unit. Reassuringly, Appendix Table A7 shows that the presence of a higher-ranked officer was not correlated with an event being classified ex-ante as high risk.

**Alternative sample composition** Appendix Table A8 demonstrates that the results are robust to changes in the estimation sample. Panel A reproduces the main effects for reference. In Panel B, we include data from blackout days. Not surprisingly, the treatment effect is still present but is smaller in magnitude since the sample includes days when officers were randomly not handed out cameras.

Panel C looks at incidents with two officers, which was the modal unit size. The results show that when we restrict the sample to these events, the effect on use of force loses precision, even though it remains negative and sizable. The effects of the Adverse Interaction Index remain strong and statistically significant. Panel D excludes incidents with more than four police officers, and the results remain virtually the same. Finally, we use data from before the experiment and, reassuringly, Panel E shows that all of our effects disappear when considering the randomization inference  $p$ -values. Overall, our results remain qualitatively unchanged in this exercise and highlights that no anticipation to treatment takes place.

**Exploiting only observational variation.** In Appendix Section B1, we estimate treatment effects exploiting only observational variation, exploring the spatially explicit dimension of our intervention. Using the group of precincts that did not participate in the experiment, we assess how use of force evolved in the experimental precincts vis-à-vis those in which no officer wore a camera. This approach can allay some concerns about potential unobservable within-precinct spillovers.

### 3.5 Welfare Analysis

We now assess the welfare implications of body-worn cameras by calculating the marginal value of public funds (MVPF), following the methodology of [Finkelstein and Hendren \(2020\)](#). The full details of our approach are provided in Appendix D and summarized below.

The results in this section indicate a significant reduction in the use of force, alongside various other effects, such as improved reporting, particularly in cases of domestic violence. Assigning monetary values to such a broad range of outcomes is inherently challenging. Our analysis will primarily focus on the effects of the prevented use of force. We argue that, due to the exclusion of other potential channels, our estimates likely underestimate the true MVPF.

To quantify the effects related to the use of force, we break down the benefits into four broad categories: *i*) police lethality – incidents involving the use of firearms that result in fatalities; *ii*) lethal weapons without lethal consequences – instances where firearms are used but do not result in fatalities; *iii*) non-lethal weapons – the use of devices such as tasers or other non-lethal tools; *iv*) physical force by the police such as the use of physical actions or restraint methods by officers.

First, we calculate the baseline levels for the four categories of use of force and express them as rates per 100,000 inhabitants per year. Next, we apply the treatment effect estimates to calculate the number of averted incidents, which are then assigned monetary values based on estimates from the literature.<sup>28</sup> Additionally, we adjust the WTP for averting non-lethal encounters used in the

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<sup>28</sup>For averted lethal incidents we use the median value of statistical life across five estimates among [Corbi et al. \(2006\)](#), [Cerqueira \(2014\)](#), [Rocha and Soares \(2015\)](#); [Viscusi and Masterman \(2017\)](#), [Cerqueira et al. \(2020\)](#) and [Pereira et al. \(2020\)](#). We quantify the benefit of other averted

literature to obtain adequate numbers for our context. (Cohen and Piquero, 2009, Williams et al., 2021). Finally, we also account for the potential effects of reduced arrests and use of handcuffs considering the median criminal court cost for the victim across a battery of crimes.

In our preferred estimate, we find that body-worn cameras (BWCs) are likely to deliver an MVPF of 9.26. This estimate is notably higher than previous evaluations of BWCs conducted in the United States.<sup>29</sup> Our findings also compare favorably to other police interventions, which changed the number of officers or police stations (Chalfin and McCrary, 2018, Facchetti, 2021). Chalfin and McCrary (2018) find that the benefit-cost ration of increased policing in the US depends on the valuation of the VSL with some estimates indicating overpolicing. In our analysis, we find an MVPF still larger than one when we consider the lowest range of VSL found in the literature.

We argue that the relatively high MVPF estimate is driven by a combination of factors that are likely to be replicated across Latin American countries. These elevated estimates stem from conditions that make BWCs particularly impactful in Latin America, and Brazil specifically. Notably, the baseline levels of police lethality in our context are significantly higher than those observed in the United States (see Figure A2). When this high baseline is paired with a substantial estimated reduction in lethality, it leads to a correspondingly large benefit. Higher MVPFs in low- or middle-income countries is also seen in other types of interventions (Hahn et al., 2024).

## 4 Reconciling BWC Effects with the Literature

The literature on BWCs' effects on use of force has produced mixed results thus far. While recent meta-studies (Lum et al., 2020; Williams et al., 2021) have failed to detect effects of BWCs, some recent evidence is more encouraging and points to detectable and meaningful effects on use of force (see, for example, Braga et al., 2018; Monteiro et al., 2022; Ferrazares, 2023; Kim, 2024). In this section, we attempt to make sense of these differences and reconcile the mixed results in the incidents using estimates from Cohen and Piquero (2009).

<sup>29</sup>Williams et al. (2021) estimated a cost-benefit ratio of 4.95 for BWC implementation in the United States.



literature.

We argue that the main difference between estimates in the literature stems from the ability to control for contamination effects.<sup>30</sup> In contrast, we are able to observe the specific incidents in which cameras were used, and assign the treatment at that level. The granular level of observation also provides for increased precision through increased sample sizes and fixed effects at fine levels.

To show this point, we reproduce aggregate the experimental setting – using exclusively officer or shift level variation, commonly used in the literature – and compare results with the treatment effects at the event level, our unit of analysis. This allows us to investigate the extent to which the BWC camera effect estimates are sensitive to the experimental design.

We start with mimicking an *officer-centric* variation, which randomizes officers into treatment and control groups. As outcome variable  $y_{od}$ , we measure the share of incidents in which an officer  $o$  used force over a time period – say, a day of the week  $d$ . We then explore the experimental variation in officer allocation to the treatment and control groups in the following specification:

$$y_{od} = \beta_{\text{officer}} \times \text{Treated Officer}_o + \eta_{bw} + \tau_d + \phi_o + \epsilon_{od} \quad (2)$$

excluding blackout days, so we solely rely on the between-officer variation. As in our main specification, we include police precinct-by-week fixed effects ( $\eta_{bw}$ ) along with day-of-the-week fixed effects ( $\tau_d$ ). We also include stratification bins fixed by officer  $o$  effects  $\phi_o$ . The disturbance  $\epsilon_{od}$  is clustered at the police precinct-by-week level.  $\text{Treated Officer}_o$  equals 1 if the officer was assigned to wear a camera. We are interested in the estimated  $\beta_{\text{officer}}$ .

The results show an attenuation: the effect size is reduced from our event-level benchmark of 61.2% to 26.5%. The attenuation is not surprising: in our design, one third of the officers were

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<sup>30</sup>Spillover and contamination effects may have attenuated some previous evaluations of BWCs which relied exclusively on officer or shift variation. The potential for spillover effects in this context is considerable, as, for example, control officers patrolling with treated officers were indirectly treated and very likely to alter their behavior.

randomly selected to wear a camera. Since a noticeable share of events attended by control group officers was, in fact, indirectly treated due to the presence of other experimental officers carrying cameras, this downward biases the treatment effect estimate since a large share of events coded as being attended by control-group officers were treated.

We then present the treatment effect estimate implied in our data carrying out the analysis when we solely exploit treatment and control variation across shifts. In this case, we collapse the data at the precinct-by-day level, and we exploit the fact that our research design allows us to contrast blackout and non-blackout days to give us treated and untreated shifts. In the following specification, the outcome variable  $y_{bd}$  is the share of events in which force was used at police precinct  $b$  during day of the week  $d$ ,

$$y_{bd} = \beta_{\text{shift}} \times \text{Treated Shift}_d + \eta_{bw} + \tau_d + \epsilon_{bd}. \quad (3)$$

The fixed effects we control for are police precinct-by-week and day-of-the-week. The error term  $\epsilon_{bd}$  is clustered at the police precinct-by-week level. The estimated effect sizes are around 16%, a substantial attenuation from the 61.2% reduction in use of force that was estimated from the event-level specification, and statistically insignificant. This effect size is comparable with studies that used variation at the shift level.

We provide further details in Appendix Section B. There we overview the existing evidence about body-worn cameras and further study how the temporal resolution of the outcomes may have affected previous evaluations. Importantly, and as alluded in Section 3.4, we leverage on a differences-in-differences design and compare to other studies in the literature that adopted similar methodologies.

## 5 Conclusion

The issue of police violence is a global problem, and there is an urgent need to find ways to increase accountability. In 2021, Brazil suffered from high levels of police brutality, resulting in

6,145 deaths caused by police action.<sup>31</sup> Through a large-scale experiment in Santa Catarina, Brazil, we have revealed that the implementation of BWCs by police can significantly lower instances of force by an average of 61.2%. While our results did not measure deaths in police action, it is conceivable that some of the benefits in reduction of use of force could translate into curbing loss of life during police encounters.

We find evidence that implies that the change in conduct is not solely on the part of the citizens, and that some behavioral change can be taking place on the part of the police. We also have revealed the impacts of the devices on improving police reporting, especially concerning domestic violence (the reporting of which grew by 69%), which goes in tandem with a reduction of officers filing empty reports. This is important, as accurate reporting by police officers has been demonstrated to improve the rate at which crimes are cleared (Blanes i Vidal and Kirchmaier, 2018).

Taken together, our results indicate that using BWCs can increase the accountability of police officers. This has important policy implications. The positive results of this research paper have already proven to impact the policymaking and the adoption of police BWCs. Following this study, the government of Santa Catarina purchased 2,425 cameras in August 2019, and now virtually all incidents are recorded in the state.<sup>32</sup> Since then, the Military Police of São Paulo state has also adopted BWCs and Monteiro et al. (2022) evaluated the effect of cameras in this context using observational and aggregated data, finding similar effects in the reduction of use of force. More recently, BWC policy has been formulated at the national level through the work of the Ministry of Justice.<sup>33</sup> The Supreme Court of Brazil mandated that the police of the state of Rio de Janeiro should adopt BWCs in the wake of allegations of excessive use of force.<sup>34</sup> More generally, BWCs

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<sup>31</sup>Source: Forum Brasileiro de Segurança Pública, <http://forumseguranca.org.br:3838/>, accessed November 2023.

<sup>32</sup>Source: <https://www.pm.sc.gov.br/noticias/policia-militar-lanca-cameras-policiais-individuais>, accessed March 22nd 2023

<sup>33</sup>Source: <https://www1.folha.uol.com.br/cotidiano/2023/02/governo-lula-prepara-programa-de-cameras-em-uniformes-policiais-para-o-1o-semester.shtml>, accessed March 22nd 2023.

<sup>34</sup>Source: <https://portal.stf.jus.br/noticias/verNoticiaDetalhe.asp?idConteudo=508510&ori=1>.

feature as a key program of the Bureau of Justice Assistance of the United States’ Department of Justice.<sup>35</sup> The growing take-up of BWCs suggests that there is a latent demand for interventions that increase scrutiny and oversight of the police; and that cameras can provide one such solution with visible impacts.

Our final point is concerned with external validity. While our study provides evidence of strong effects of BWCs, it was also conducted under experimental conditions with relatively strong involvement of police supervisors. Other successful implementations, such as in São Paulo, also stressed the role of supervisors in ensuring that street-level officers, who directly interact with the public, adhere to camera-use mandates and protocols.

This suggests that BWCs should be seen as part of a broader set of incentives set out by the institutions, and any understanding of BWCs’ effects should account for how monitoring devices interplay with these incentives. The role of monitoring in the workplace as a broader topic has been stressed both in the context of private- and public-sector agents (see [Bandiera et al. \(2010\)](#), [Fredriksen et al. \(2020\)](#), and [Fenizia \(2022\)](#), among others). This may corroborate the view that strong supervisions may be necessary in order for similar effects to materialize in future implementations. This further highlights that advancing our understanding about the importance of monitoring becomes more fundamental when the outcomes in question are police violence, use of force, and, in extreme cases, the death of citizens resulting from these actions.

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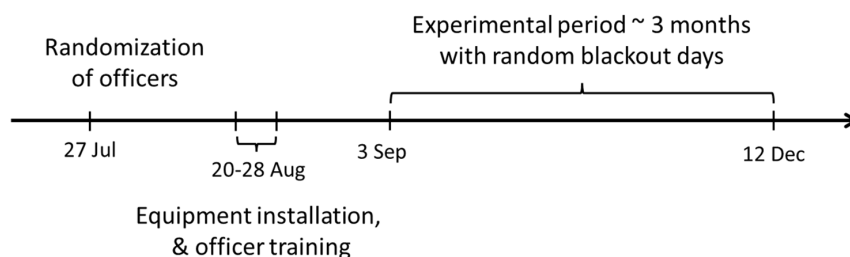
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# Tables and Figures

**Figure 1: Timeline and experimental design**

**Panel A. Experimental timeline**

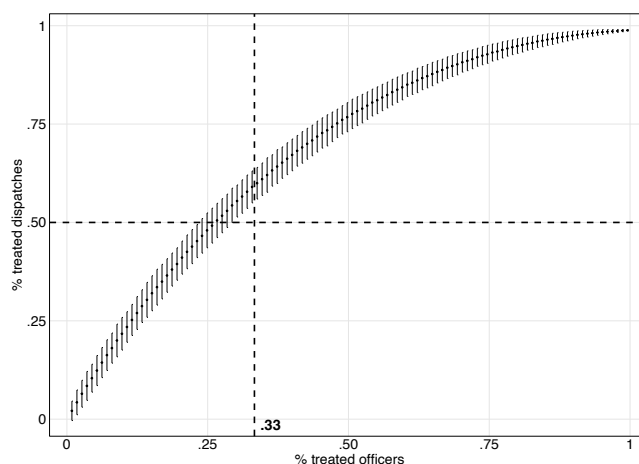


**Panel B. Embedding variation into the police incident data**

*Pre-intervention baseline data...*

*... was used to induce treatment allocation at the event level*

| Incident | Off. 1 | Off. 2 | Treated |         |              |
|----------|--------|--------|---------|---------|--------------|
|          |        |        | A       | A and B | Shift        |
| 1        | A      | B      | 1       | 1       | 1 (Regular)  |
| 2        | A      | C      | 1       | 1       | 1 (Regular)  |
| 3        | A      | D      | 1       | 1       | 1 (Regular)  |
| 4        | B      | C      | 0       | 1       | 1 (Regular)  |
| 5        | B      | D      | 0       | 1       | 1 (Regular)  |
| 6        | C      | D      | 0       | 0       | 1 (Regular)  |
| ⋮        |        |        |         |         |              |
| 25       | A      | B      | 1       | 1       | 0 (Blackout) |
| 26       | A      | C      | 1       | 1       | 0 (Blackout) |
| 27       | A      | D      | 1       | 1       | 0 (Blackout) |
| 28       | B      | C      | 0       | 1       | 0 (Blackout) |
| 29       | B      | D      | 0       | 1       | 0 (Blackout) |
| 30       | C      | D      | 0       | 0       | 0 (Blackout) |



*Notes:* This figure presents the experimental timeline. Panel A provides the timeline of the experiment that was conducted in 2018. The table in Panel B illustrates that a sparse (i.e., less than half) camera allocation to officers is required to induce a 50% treatment – 50% control variation at the incident level. The figure in Panel B shows the exact simulations that were conducted with actual police pre-intervention incident data. It shows that optimal treatment allocation would be achieved by allocating cameras to between one fourth and one third of police officers.

**Table 1: Effects of BWCs on reporting**

|                                                                              | Event<br>Recorded                 | Event<br>Registered<br>with No Info | Verbal<br>Attrition/<br>Threat    | Noise<br>Complaint             | Burglary                        | Assault                         | Domestic<br>Violence             | Share of<br>Reports with<br>Victims | Generated<br>Investigative<br>Report |
|------------------------------------------------------------------------------|-----------------------------------|-------------------------------------|-----------------------------------|--------------------------------|---------------------------------|---------------------------------|----------------------------------|-------------------------------------|--------------------------------------|
|                                                                              | (1)                               | (2)                                 | (3)                               | (4)                            | (5)                             | (6)                             | (7)                              | (8)                                 | (9)                                  |
| <b>Panel A. Main Effects</b>                                                 |                                   |                                     |                                   |                                |                                 |                                 |                                  |                                     |                                      |
| Treated Event                                                                | 24.043***<br>(1.873)<br>p = 0.000 | -2.770**<br>(1.239)<br>p = 0.164    | 0.237<br>(0.645)<br>p = 0.774     | 0.126<br>(0.550)<br>p = 0.870  | 0.842*<br>(0.427)<br>p = 0.131  | 0.709*<br>(0.388)<br>p = 0.096  | 1.138***<br>(0.351)<br>p = 0.000 | 2.783***<br>(0.805)<br>p = 0.004    | 3.101**<br>(1.190)<br>p = 0.060      |
| <b>Panel B. Heterogeneity by Officer Rank</b>                                |                                   |                                     |                                   |                                |                                 |                                 |                                  |                                     |                                      |
| Treated Event by Officer(s) with Soldier Rank                                | 24.974***<br>(2.040)<br>p = 0.000 | -2.840**<br>(1.329)<br>p = 0.147    | 1.130*<br>(0.672)<br>p = 0.200    | -0.047<br>(0.608)<br>p = 0.953 | 1.106**<br>(0.478)<br>p = 0.065 | 0.761*<br>(0.393)<br>p = 0.099  | 1.269***<br>(0.364)<br>p = 0.001 | 3.240***<br>(0.871)<br>p = 0.004    | 3.247**<br>(1.289)<br>p = 0.049      |
| Treated Event by Officer(s) with Higher-than-Soldier Rank                    | 18.906***<br>(2.244)<br>p = 0.000 | -2.230<br>(2.417)<br>p = 0.634      | -3.073***<br>(1.103)<br>p = 0.037 | 0.528<br>(1.173)<br>p = 0.693  | -0.164<br>(0.730)<br>p = 0.883  | 0.699<br>(0.725)<br>p = 0.394   | 0.619<br>(0.616)<br>p = 0.273    | 1.503<br>(1.349)<br>p = 0.391       | 2.201<br>(2.077)<br>p = 0.609        |
| Treated Event by Officers of Both Types                                      | 24.788***<br>(2.678)<br>p = 0.000 | -3.524<br>(3.054)<br>p = 0.426      | -3.675***<br>(1.214)<br>p = 0.022 | 1.503<br>(1.260)<br>p = 0.371  | -0.217<br>(1.102)<br>p = 0.848  | 0.068<br>(1.200)<br>p = 0.947   | 0.697<br>(0.722)<br>p = 0.341    | -0.108<br>(1.880)<br>p = 0.958      | 3.957<br>(3.217)<br>p = 0.394        |
| <b>Panel C. Heterogeneity by Treatment Intensity</b>                         |                                   |                                     |                                   |                                |                                 |                                 |                                  |                                     |                                      |
| Treated Event by 1 Camera                                                    | 22.430***<br>(1.881)<br>p = 0.000 | -2.626**<br>(1.272)<br>p = 0.186    | 0.143<br>(0.642)<br>p = 0.869     | 0.520<br>(0.559)<br>p = 0.522  | 1.045**<br>(0.454)<br>p = 0.066 | 0.427<br>(0.384)<br>p = 0.321   | 1.038***<br>(0.342)<br>p = 0.004 | 2.485***<br>(0.875)<br>p = 0.014    | 2.825**<br>(1.275)<br>p = 0.082      |
| Treated Event by 2 or More Cameras                                           | 30.473***<br>(2.632)<br>p = 0.000 | -3.345*<br>(1.890)<br>p = 0.238     | 0.609<br>(1.144)<br>p = 0.617     | -1.446<br>(0.947)<br>p = 0.265 | 0.034<br>(0.616)<br>p = 0.968   | 1.834**<br>(0.784)<br>p = 0.007 | 1.535**<br>(0.655)<br>p = 0.007  | 3.972***<br>(1.184)<br>p = 0.007    | 4.201**<br>(1.599)<br>p = 0.098      |
| p-value for $\beta_{\text{Soldier}} \neq \beta_{\text{Higher than Soldier}}$ | 0.006                             | 0.807                               | 0.000                             | 0.657                          | 0.122                           | 0.930                           | 0.267                            | 0.214                               | 0.625                                |
| p-value for $\beta_1^{\text{Camera}} \neq \beta_2^{\text{or More Cameras}}$  | 0.000                             | 0.680                               | 0.657                             | 0.027                          | 0.097                           | 0.066                           | 0.406                            | 0.239                               | 0.390                                |
| Mean Dep. Var.                                                               | 0.000                             | 47.268                              | 9.940                             | 8.661                          | 4.787                           | 3.618                           | 1.644                            | 13.832                              | 32.761                               |
| N                                                                            | 13274                             | 13274                               | 13274                             | 13274                          | 13274                           | 13274                           | 13274                            | 13274                               | 13274                                |

*Notes:* This table documents the effects of BWCs on recording and reporting measures, including criminal typology. The dependent variables are “event recorded”, indicating that the incident was partially or fully recorded using the BWC; “event registered with no info”, indicating no criminal typology was recorded; the five most frequent criminal typologies reported: “verbal attrition/threat”, “noise complaint”, “burglary”, “assault” and “domestic violence”; and “share of reports with victims” and “generated investigative report”, when officers reported events to the Civil Police, who would proceed with investigations. Panel A presents the main results capturing the average intent-to-treat effect. Panel B explores rank heterogeneity of who wore the camera. Panel C investigates treatment intensity heterogeneity based on the number of officers wearing a camera in events. All dependent variables are multiplied by 100. Specifications include police precinct-by-week, day of the week, number of officers and stratification bins fixed effects. Shifts without a camera are excluded from the regression, which follows specification (1). Standard errors are clustered at the precinct-by-week level. \*\*\* p<0.01; \*\* p<0.05; \* p<0.1. The randomization inference p-values are indicated below the standard errors. The test for equality of the heterogeneous treatment effects is performed using the delta method.

**Table 2: Effects of body-worn cameras on use of force**

|                                                                                            | Use of<br>Force                   | Handcuff<br>and/or<br>Arrest   | Contempt,<br>Resistance<br>and/or Dis-<br>obedience | Adverse<br>Interac-<br>tion<br>Index |
|--------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------|-----------------------------------------------------|--------------------------------------|
|                                                                                            | (1)                               | (2)                            | (3)                                                 | (4)                                  |
| <b>Panel A. Main Effects</b>                                                               |                                   |                                |                                                     |                                      |
| Treated Event                                                                              | -0.425***<br>(0.157)<br>p = 0.009 | -0.320<br>(0.471)<br>p = 0.584 | -0.263<br>(0.196)<br>p = 0.280                      | -0.371**<br>(0.149)<br>p = 0.030     |
| <b>Panel B. Heterogeneity by Officer Rank</b>                                              |                                   |                                |                                                     |                                      |
| Treated Event by Officer(s) with Soldier Rank                                              | -0.471***<br>(0.166)<br>p = 0.007 | -0.294<br>(0.504)<br>p = 0.619 | -0.390*<br>(0.219)<br>p = 0.121                     | -0.444***<br>(0.161)<br>p = 0.015    |
| Treated Event by Officer(s) with Higher-than-Soldier Rank                                  | -0.304<br>(0.325)<br>p = 0.311    | -0.520<br>(1.156)<br>p = 0.734 | 0.095<br>(0.413)<br>p = 0.871                       | -0.174<br>(0.311)<br>p = 0.606       |
| Treated Event by Officers of Both Types                                                    | -0.065<br>(0.541)<br>p = 0.909    | -0.001<br>(1.707)<br>p = 1.000 | 0.604<br>(0.641)<br>p = 0.522                       | 0.154<br>(0.502)<br>p = 0.813        |
| <b>Panel C. Heterogeneity by Treatment Intensity</b>                                       |                                   |                                |                                                     |                                      |
| Treated Event by 1 Camera                                                                  | -0.392***<br>(0.143)<br>p = 0.017 | -0.091<br>(0.533)<br>p = 0.884 | -0.207<br>(0.205)<br>p = 0.390                      | -0.330**<br>(0.145)<br>p = 0.048     |
| Treated Event by 2 or More Cameras                                                         | -0.554*<br>(0.298)<br>p = 0.083   | -1.233<br>(0.804)<br>p = 0.209 | -0.487<br>(0.357)<br>p = 0.293                      | -0.535**<br>(0.265)<br>p = 0.111     |
| <b>Panel D. Heterogeneity by Ex-ante Event Risk Assessment</b>                             |                                   |                                |                                                     |                                      |
| Treated Event x Low Risk                                                                   | -0.414**<br>(0.161)<br>p = 0.004  | -0.433<br>(0.501)<br>p = 0.445 | -0.381*<br>(0.216)<br>p = 0.095                     | -0.403**<br>(0.160)<br>p = 0.010     |
| Treated Event x High Risk                                                                  | -0.489<br>(0.703)<br>p = 0.452    | 0.718<br>(1.474)<br>p = 0.636  | 0.777<br>(0.811)<br>p = 0.487                       | -0.070<br>(0.640)<br>p = 0.928       |
| <b>Panel E. Heterogeneity by Baseline Use of force</b>                                     |                                   |                                |                                                     |                                      |
| Treated Event x Below Median Use-of-Force                                                  | -0.265**<br>(0.133)<br>p = 0.079  | -0.265<br>(0.517)<br>p = 0.675 | -0.087<br>(0.170)<br>p = 0.714                      | -0.206*<br>(0.123)<br>p = 0.189      |
| Treated Event x Above Median Use-of-Force                                                  | -1.059**<br>(0.436)<br>p = 0.006  | -0.527<br>(0.827)<br>p = 0.590 | -0.962*<br>(0.541)<br>p = 0.092                     | -1.025**<br>(0.402)<br>p = 0.009     |
| $p$ -value for $\beta^{\text{Soldier}} \neq \beta^{\text{Higher than Soldier}}$            | 0.641                             | 0.854                          | 0.309                                               | 0.442                                |
| $p$ -value for $\beta^{\text{1 Camera}} \neq \beta^{\text{2 or More Cameras}}$             | 0.507                             | 0.221                          | 0.436                                               | 0.386                                |
| $p$ -value for $\beta^{\text{Low Risk}} \neq \beta^{\text{High Risk}}$                     | 0.919                             | 0.459                          | 0.185                                               | 0.623                                |
| $p$ -value for $\beta^{\text{Below Median Use of Force}} \neq \beta^{\text{Above Median}}$ | 0.057                             | 0.770                          | 0.094                                               | 0.029                                |
| Mean Dep. Var. Control Events                                                              | 0.694                             | 5.427                          | 0.932                                               | 0.790                                |
| $N$                                                                                        | 13274                             | 13274                          | 13274                                               | 13274                                |

*Notes:* This table shows the effect of BWCs on use of force. The dependent variables are “use of force”, which indicates if there was any deployment of physical, non-lethal (mechanical), or lethal force by the police, not considering use of handcuff or arrest; “arrest and/or the use of handcuffs”, which is an indicator for if handcuffs were used or if any arrests made; “contempt, resist, and/or disobey” which is an indicator for if charges of contempt, disobedience, or resistance toward the police were registered; “Adverse Interaction Index” is the standardized inverse-covariance weighted average of the three indicators in the group. Panel A presents the main results capturing the average intent-to-treat effect. Panel B explores rank heterogeneity of who wore the camera. Panel C investigates treatment intensity heterogeneity based on the number of officers wearing a camera in events. Panel D explores heterogeneity by the ex-ante risk level of the events, which characterizes an event as low risk if it had no weapons on the scene, if there were no injuries, if the suspect was not on site and if there was no material risk of general unrest. Panel E explores the heterogeneity by baseline use of force in areas of the municipalities. All dependent variables are multiplied by 100. Specifications include police precinct-by-week, day of the week, number of officers and stratification bins fixed effects. Shifts without camera were excluded from the regression. Standard errors are clustered at the precinct-by-week level. \*\*\*  $p < 0.01$ ; \*\*  $p < 0.05$ ; \*  $p < 0.1$ . The randomization inference  $p$ -values are indicated below the standard errors. The test for equality of the heterogeneous treatment effects is performed using the delta method.

**Table 3:** Testing for endogenous allocation of BWC to Events

|                | High<br>Baseline<br>Use of<br>Force | Latitude                      | Longitude                       | High<br>Baseline<br>Income    | Time to<br>Dispatch<br>(Minutes) | Time to<br>Dispatch<br>Greater<br>than 5<br>min. | Active<br>Policing             | Telephone<br>Initiated<br>Incident | High<br>Ex-Ante<br>Risk        |
|----------------|-------------------------------------|-------------------------------|---------------------------------|-------------------------------|----------------------------------|--------------------------------------------------|--------------------------------|------------------------------------|--------------------------------|
|                | (1)                                 | (2)                           | (3)                             | (4)                           | (5)                              | (6)                                              | (7)                            | (8)                                | (9)                            |
| Treated Event  | -0.170<br>(0.896)<br>p = 0.869      | 0.001<br>(0.001)<br>p = 0.708 | 0.002**<br>(0.001)<br>p = 0.374 | 1.726<br>(1.313)<br>p = 0.541 | -1.719<br>(1.269)<br>p = 0.223   | -1.348<br>(1.100)<br>p = 0.275                   | -0.201<br>(0.573)<br>p = 0.838 | 0.235<br>(0.574)<br>p = 0.812      | -0.341<br>(0.587)<br>p = 0.586 |
| Mean Dep. Var. | 20.080                              | -27.468                       | -48.787                         | 48.639                        | 10.701                           | 43.868                                           | 7.637                          | 92.180                             | 10.378                         |
| Control Events | 13274                               | 13274                         | 13274                           | 13274                         | 13274                            | 13274                                            | 13274                          | 13274                              | 13274                          |
| N              |                                     |                               |                                 |                               |                                  |                                                  |                                |                                    |                                |

*Notes:* This table presents tests for the characteristics of the incident that could suggest endogenous allocation with respect to treatment assignment. The dependent variables are: high baseline use of force, which indicates whether the event happened in census tracts with an above median baseline use of force; latitude and longitude, both measured in degrees; high baseline income, which indicates whether the event happened in census tracts with an above median income in the baseline; time to dispatch, a measure of the interval between communication and arrival in minutes and a dummy to whether this interval was higher than five minutes; active policing, a dummy indicating if police self-initiated the event; telephone-initiated incident, which is an indicator for if the event was communicated through the telephone central; and high ex-ante risk, which is a measure of ex-ante risk that characterizes an event as low risk if it had no weapons on the scene, if there were no injuries, if the suspect was not on site, and if there was no material risk of general unrest. Sample includes all events in the experimental period and excluded blackout shifts. Specification includes police precinct-by-week, day of the week, number of officers and stratification bins fixed effects. Standard errors are clustered at the precinct-by-week level. \*\*\* p<0.01; \*\* p<0.05; \* p<0.1. The randomization inference p-values are indicated below the standard errors.

## Online Appendix

# “Monitoring technology: the impact of body-worn cameras on citizen-police interactions”

For Online Publication

Barbosa, Fetzer, Soto-Vieira, and Souza

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## A Implementation Details

**Preparations and randomization** We obtained the full roster of citizen-facing police officers in the beginning of July 2018. Officers without citizen-facing duties, such as those with administrative roles, were not considered eligible for camera use. We used pre-intervention incident data to validate if the list that was sent to us had contained all citizen-facing police officers. We then confirmed that there was no self-selection of officers into the study sample.

During this period, all tests with the cameras and docking stations were conducted to ensure that the information necessary for the experiment was correct and to minimize technical issues during the experiment period. Prior to the start of the experiment, all officers were briefed how to use the equipment, and how to adjust standard operating procedures allowing for the use of a camera (in particular, it was made clear that officers were required to verbally communicate to citizens that the events were being recorded). Importantly, *all* officers were briefed — irrespective of their treatment status — to avoid the briefing itself confounding the BWC treatment effects.

The implementation timeline is depicted in the Figure 1 above. We randomized officers and blackout days on July 7, 2018. Shift-level treatment allocation was randomized before the start of the experiment, but we only communicated to the police precincts on the preceding evening through dedicated WhatsApp groups established for this purpose. This was supposed to avoid the potential for the endogenous selection of any aspect of the policing activity with respect to the anticipation of blackout days. Importantly, the blackout applied to officers starting their shifts. That is, officers already on duty at midnight of the start of the blackout would continue to use their cameras until the end of their shift; conversely, any shift that started during blackout, even if it continued beyond the blackout, would not be recorded. This feature was necessary for logistic reasons: the police deemed it not practical or desirable to interfere in the apparatus of the incident units after they had left the precinct headquarters.

**Intervention step by step** Once the experiment period started, the intervention progressed as follows. At the start of their shifts, treated officers would obtain their camera, from the armory



section of the police precincts — from where they obtained their gun, radio, and other equipment for regular and special use. The armory sections are usually very secluded and considered to be a high-security environment due to the nature of the material that is stored therein, and only a few high-ranking officers have access to those rooms. Importantly, the docking stations, which both downloaded the videos at the end of every shift and recharged the cameras, were located inside the armory rooms. This ensured that not only the equipment was maintained, regularly inspected, and kept in good working order throughout the experiment, but also ensured that docking stations and cameras themselves were not interfered with or violated during the experiment.

The docking stations were remotely accessible from the PMSC headquarters. Videos were stored locally for 30 days and transferred to the central HQ on demand due to bandwidth issues. The research team established routines to consolidate the cameras' automatic logs in a central database. In this way, it was possible to observe if a given incident generated a video recording, as well as the corresponding docking station and filenames. After finishing their shift, police officers would hand back their cameras to the armory section, which would then be docked in the station and readied for their next use. This recycling process usually took between four and six hours for a full battery charge that lasted at least eight hours in continuous regular use.

On the preceding night before control shifts, the research team would message the officers responsible for the armory sections in each police precinct telling them to **not** give cameras to treated officers. So the officers who started their shifts on the blackout day would receive from the armory all their equipment but the cameras.

On any given day, units would be composed by on average two officers. If any of those was assigned to wear a camera at the officer-level randomization, this incident (as well as the events they tended to) was classified as a treatment one. Thus, the average treatment effect of BWC implementation over police activity and police-citizen interactions was identified by comparing events attended by incidents with at least one officer assigned to wear a camera with events with no officers with a camera.

As for blackout shifts, all treated officers were not allowed to wear cameras. Therefore, we

can compare events attended by treated incidents with events attended by control incidents in days in which no treatment officer was allowed to wear cameras, allowing us to identify if the effects persisted when the treatment technology was not present. Importantly, the dispatch operators were blind to whether units were manned with officers wearing a BWC. This prevented the endogenous allocation of calls to be recorded (or, conversely, to avoid being recorded).

## **B The BWC Literature and Use of Force**

Experiments on the effects of BWCs on use of force do not consistently show that cameras effectively work to decrease excessive use of force, as shown by [Lum et al. \(2020\)](#) and [Williams et al. \(2021\)](#). Appendix Table [A9](#) lists the main BWC papers in the literature.<sup>1</sup> We include their main features, e.g., the number of citations, country, sample size, share of treated units, and whether any effect of use of force was detected. As can be seen, the literature is not conclusive on BWCs' effects on use of force. We argue below that most of the experiments with BWCs were potentially affected by methodological issues which attenuated the cameras' effects.

We start in Panel A with the studies that allocated cameras on the basis of shifts. All of the seven papers listed were RCTs, and they allocated cameras on the basis of treatment and control shifts. We argue that this design is potentially problematic, as a given officer may be allocated to both a treatment and a control shift over time. This may be an important SUTVA assumption violation if, for example, officers alter their behavior after using a camera, e.g., through learning, or if there are across-officers spillover effects ([Ariel et al., 2017](#)). Out of the seven studies that used a shift analysis, five had use of force as an outcome and only one found statistically significant results (at the 5% level) that suggest that BWCs affect use of force. [Ariel et al. \(2015\)](#) conducted the first experiment on BWCs and it is by far the most cited paper in the literature. The shifts were randomized to be conducted with and without cameras, and the results suggest that BWCs reduce use of force by the police. However, these effects were barely significant at the 10% level.

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<sup>1</sup>This is not intended as a literature review but a selective and partial read on the studies that we found to be most prominent in the literature.

Following that, [Ariel et al. \(2016b\)](#) repeated the same design across multiple sites, and the results show null effects of BWCs on use of force. [Ariel et al. \(2016a\)](#) suggested that one potential explanation for these muted results stemmed from compliance with the protocol. They showed that use-of-force rates were higher in sites where the compliance with the protocol was lower, and vice-versa. [Magaloni \(2019\)](#) did not find any effects of BWCs on use of force, and the experiment faced issues with low compliance as well. With an experiment in the UK, [Henstock and Ariel \(2017\)](#) used shift randomization and found that BWC were effective in reducing use of force, in particular physical restraint and non-compliant handcuffing.

We move to officer-centric designs in Panel B. The literature shifted to officer-level allocation to ensure officers were always in the same assigned group throughout the duration of the experiment. This design also presents its challenges. First, contamination is a substantial concern: among the officer-centric papers we could identify, 60% routinely had more than one police officer per incident, which can mechanically result in contamination between officers if both a treated and a control officer are in the same incident. Moreover, all officer-level studies treated half of the police officers, which resulted in a much higher share of treated events — if an event was considered as treated if one or more cameras were present — given that most incidents are tended by more than one police officer. In our data, simulations show that treating half of the officers would lead to 75% of the events being treated (see Panel B of Figure 1), leading to a considerably smaller control group and potentially undermining the statistical power. A study by [Braga et al. \(2020\)](#), who used officer-level randomization combined with spatial selection of districts, indeed shows evidence of large contamination from treated officers to control officers. Finally, some papers only included in the experiment officers that volunteered to wear a BWC ([Jennings et al., 2015](#); [Ready and Young, 2015](#); [White et al., 2017](#); [Jennings et al., 2017](#); [Headley et al., 2017](#); [Braga et al., 2018](#)). This can introduce self-selection bias and compromise the identification of the effects. Taken together, these design characteristics can result in muted estimated effects of BWCs on police operations.

Finally, Panel C listed papers that used of spatially explicit empirical designs. Out of 12 studies, only four looked at use of force as an outcome, and only [Kim \(2024\)](#) and [Ferrazares \(2023\)](#) found

evidence of the impact of BWCs. [Kim \(2024\)](#) used the variation in the timing of the adoption across US agencies to assess the effects of BWCs on a national level using a difference-in-differences with two-way fixed effects. While this strategy does not have to deal with the spillover that can occur between officers, it relies on the strong identifying assumption that adoption timing is independent of agency characteristics. Similarly, [Miller and Chillar \(2021\)](#) explored the staggered adoption of BWCs to study the effects on fatalities that arise from citizen-police interactions. While this is not the typical use-of-force outcome considered by us and the literature, it represents an extreme and infrequent case of when escalation unfolds. [Bollman \(2021\)](#) studied the effects of BWCs on court outcomes, also using a spatially explicit difference-in-difference. She found a significant reduction in new case filings for offenses initiated during a citizen-police interaction, suggesting an improvement in these encounters.

Overall, some papers in this panel did not follow rigorous program evaluation techniques, and some did not even perform statistical inference methods. Nonetheless, meta-analyses of the existing studies have found no statistically significant effect of BWCs on use of force, even though the point estimate is negative ([Lum et al., 2020](#); [Williams et al., 2021](#)).

## **B.1. Reconciling BWC Effects with the Literature**

As seen above, we have classified the papers as shift-centric-, officer-centric or spatially explicit according to the experimental design and evaluation and measurement strategy adopted for the implementation of BWCs. While most shift- and officer-centric papers are RCTs, the majority of papers that study the implementation of BWCs across space use difference-in-differences methods. As argued, our paper is unique since we could study the effects of BWCs using the three possible available designs in the literature. Moreover, the granularity of our data and the variation across officers, shifts, and space allows us to replicate the evaluation designs used in most past studies *in our data*. Our study directly nested shift- and officer-centric research designs, and we also could leverage spatial variation in the BWC implementation.

We first focus on specific dimensions of the randomization. Doing so allows us to reproduce

the experimental setting of other studies by reanalyzing the data at the officer or shift levels.<sup>2</sup> The level of detail of our dataset and the two-layer design of our experiment allows us to replicate shift- and officer-centric designs in our data.

In Panel A of Figure A6, we present the results of the comparison explained in Section 4 in a visual format. The point estimate displayed in red shows the 61.2% reduction in use of force between treated and untreated events, estimated from the nominal effect size of Table 2, along with the 95% confidence interval.<sup>3</sup> The event-level estimate is also indicated with the red dashed horizontal line across all panels for ease of comparison with other designs.

As noted, the results of officer-centric designs suggest an attenuation: the estimated treatment effect size capturing a decline in use of force of 26.5% is 57% smaller compared to the estimated treatment effect when carrying out the analysis at the event level. Such contamination-induced attenuation bias may affect many existing studies designed at the officer level that use difference-in-means econometric frameworks. Almost all existing studies could not directly test or measure contamination due to a lack of detailed event-level data.<sup>4</sup> Significant contamination is also observed in shift-centric designs.

Furthermore, the extent of contamination-induced attenuation bias is likely increasing in the share of officers that wear a camera. In Column (10) of Table A9, we see that virtually all officer-centric studies opted for a design with around 50% of officers assigned to wear a BWC. Assuming a similar incident unit composition as in our context, this implies that 75% of all events were treated with at least one camera (see Panel B of Figure 1), undermining the statistical power and

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<sup>2</sup>Table A9 organizes 30 papers that were surveyed. Out of those, 16 papers are RCTs, and they either randomize at the shift level or at the officer level.

<sup>3</sup>We normalize our coefficients in percentage reductions relative to the baseline incidence of use of force to render the estimates comparable across studies.

<sup>4</sup>Of the 12 studies that opted for an officer-centric research design, we were able to identify from the papers whether officers were dispatched in teams for only six studies — out of those, 66% reported that officers were dispatched in groups of two or more officers. These studies may thus have been vulnerable to such attenuation bias.

downward-biasing the treatment effect estimate when considering officer-level data. Therefore, the attenuation of the results is consistent with spillover effects since the analysis at the officer level did not account for the fact that control officers would sometimes mechanically tend to incidents with treated officers.

**Temporal Resolution of Outcome Measurement.** Accounting for unobserved time effects may be important as well. Our design at the event level allows us to control for granular time effects that might confound the effects. When comparing to the literature and focusing on the officer-level variation to uncover BWC effects, we aggregate the data either at officer by day level, officer by month level, or pooling officer observations during the whole experimental period. Specifications at coarser levels may introduce a broad range of biases, as they would imply that we cannot control for the potential confounding effect of time fixed effects which are likely very relevant. Only a few studies have considered time fixed effects as additional control variables in their respective econometric framework, with the majority of studies either ignoring time, reducing the time-dimension to before-and-after comparisons, or simply estimating differences-in-means without control variables (see Appendix Table A9).

Panel B of Figure A6 documents what happens to our point estimates with various data aggregations. The first effect size — outcomes at the officer-day level — is replicated from Equation (2), the most granular aggregation of the event-level data when exploiting our experimental variation at this unit of analysis. The second model aggregates the data to the officer-month level. In this case, the outcome variable is the share of the events with use of force by police officer  $o$  during month  $m$ . We estimate the following specification:

$$y_{om} = \beta_{\text{officer-month}} \times \text{Treated Officer}_o + \eta_{bm} + \phi_o + \epsilon_{om} \quad (4)$$

where index  $o$  refers to an officer, while index  $m$  indicates the month. We include police precinct-by-month and stratification bin by officer  $o$  fixed effects. Although the effect size does not change considerably, the precision decreases substantially. This can have two main reasons: first, naturally,

we have a smaller sample size, which implies that with conventional inference, the standard error estimates are less precise.<sup>5</sup> Furthermore, a coarser design does not allow for the inclusion of other relevant controls, such as granular fixed effects which, while uncorrelated with the treatment, in our experimental setup would improve the precision of the point estimates.

We further aggregate the data for each officer and consider the whole experimental period. Such simple group comparisons are often found in the BWC literature, accounting for at least one third of the existing experimental and non-experimental studies we surveyed. We only exploit cross-sectional variation arising from the randomization of the treatment status. We refer to this as the “pooled” specification. The estimating equation is:

$$y_o = \beta_{\text{officer-pooled}} \times \text{Treated Officer}_o + \eta_b + \phi_o + \epsilon_o \quad (5)$$

where we include only precinct and stratification bin fixed effects. Standard errors are heteroskedastic robust. The effect size from this equation is smaller in magnitude and also statistically insignificant. As a reference from the literature, [Yokum et al. \(2019\)](#) also used data at the officer level pooled during their experimental period. The effect size they found for use of force was virtually zero in both magnitude and statistical significance. Our pooled result is comparable to theirs, and, again, their point estimates shown in the horizontal black dashed line fall within our confidence intervals.

**Difference-in-Differences Designs.** In this section, we use difference-in-differences (DiD) frameworks with our data. DiD empirical frameworks are widespread in observational studies of BWC effects and typically come in two forms: either to study treated and untreated officers or treated and untreated spatial units over time. In the officer-centric DiD evaluation framework, the main concerns that may cause biased estimates are spillovers from control officers working with treatment officers and measurement error. In the spatially explicit DiD design, the prime concern is

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<sup>5</sup>Clustering the data at the precinct-month level would not be adequate due to the low number of groups that this combination provides, so we instead use heteroskedastic-robust standard errors.

statistical power, especially in the context of low compliance and coarse outcome measurement.

**Officer-centric DiD design** We first present point estimates that emerge in our data when employing a DiD design that compares the changes in outcomes associated with officers assigned to wear BWCs with those officers who never wear cameras. We estimate the following specification:

$$y_{obd} = \beta_{\text{officer-did}} \times \text{Treated Officer}_o \cdot \text{Post}_t + \eta_{bw} + \tau_d + \phi_o + \epsilon_{bdw} \quad (6)$$

where  $b$  is the police precinct,  $d$  is the day of the week, and  $w$  is the week of intervention. We include police precinct-by-week ( $\eta_{bw}$ ), day-of-the-week ( $\tau_d$ ) and stratification bins fixed effects ( $\phi_o$ ). The disturbance term is clustered at the precinct-by-week level.

The first estimate in Panel C of Figure A6 presents the treatment effect estimate, which suggests that among officers assigned to wear BWCs, use of force decreased by 32.5% relative to untreated officers. This point estimate is still around 47% smaller in absolute magnitude compared to the point estimate obtained when carrying out the analysis at the event level, and it remains consistent with the difference-in-means presented in Panel B. This is explained by the confounding effect of spillovers arising from treatment and control officers being dispatched together. Panel C illustrates the point estimate from Braga et al. (2018), which was one of the few existing studies that opted for such an evaluation approach and found smaller but statistically significant treatment effects of BWC reducing use of force.

**Spatially explicit design** We then move to a spatially explicit design where we compare outcomes of experimental against non-experimental police precincts. In our study, we obtained data from non-experimental precincts that were included in parallel research on the effects of community policing (see Blair et al., 2021; Barbosa et al., 2022). We leverage the data from those study sites to estimate a DiD design using non-experimental precincts as the control group. Naturally, these estimates may also suffer from some attenuation due to the sparsity of the treatment: only 58% of the events in experimental municipalities in the post-period were treated as per our randomization protocol. Such attenuation bias in treatment effect estimates would not arise if all officers



in treated precincts were given BWCs, as is common practice in some existing studies leveraging spatially explicit research designs. We aggregate our main outcome variables measured at the event level to the precinct-by-day level by calculating the share of events in a given precinct and day that involved use of force. We estimate the following equation:

$$y_{bdw} = \beta_{\text{precinct-did}} \times \text{Treated Precinct}_b \cdot \text{Post}_t + \eta_b + \eta_w + \tau_d + \epsilon_{bdw} \quad (7)$$

where  $b$  stands for police precinct,  $d$  is the day of the week, and  $w$  is the week of intervention. Standard errors  $\epsilon_{bdw}$  are clustered by precinct-week.

The second point estimate in Panel C of Figure A6 presents the results. We find that treated precincts presented a 0.17-percentage-point reduction in use of force, equivalent to a 46.1% decline of average use of force.<sup>6</sup> This treatment effect estimate is imprecisely estimated, suggesting that the research design may struggle with power and measurement error introduced with the aggregation, and that a large share of events in “treated” precincts were untreated. Nevertheless, the point estimate gets closest to the event-level estimate, just around 25% smaller in absolute value. Out of 12 spatially explicit studies, most often exploiting non-experimental variation, only Kim (2024) and Ferrazares (2023) found a negative treatment effect suggesting that BWCs may reduce use of force — albeit with less of an effect than what we document here.

Furthermore, we also consider this exercise as a robustness check to the treatment effect estimates, given they are obtained solely by exploiting observational variation. It also serves as corroborating evidence for the estimates presented in Section 3.

**De-policing hypothesis** We use the spatially explicit design with untreated precincts to test if the adoption of BWCs could have negative effects on police activity. One could argue that BWCs would make officers change behavior and decrease oversight, which would in turn affect the number of events registered in the police database. If the adoption of BWCs had an effect of making officers overlook crime, we would expect that the number of officer-initiated

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<sup>6</sup>Appendix Figure A7 shows that pre-trends were absent for both DiD designs in this section.

events would decrease in precincts that adopted BWCs. Relatedly, one could also argue that a decrease in police activity could increase total crime, which would in turn increase the number of telephone-initiated events.

We run a DiD in a specification similar to equation 7, but using the number of events, the number of telephone-initiated events and the number of police-initiated events as dependent variables. Table A10 shows no effects of BWC adoption on police activity in any of the three dependent variables. These results rule out the hypothesis that there was de-policing due to bodycam adoption.

## C PAP Registration

**Submission history.** Our initial study design was pre-registered on the “Evidence in Governance and Politics” (EGAP) repository as part of a broader project on the Metaketa IV round of funding that analyzed the effects related to community policing (Blair et al., 2021; Barbosa et al., 2022). Later in 2020, the EGAP repository fully migrated with the OSF repository and can now be accessed through the link <https://osf.io/yzpva/>. File dates in the OSF system refer to the migration date, not the original date we submitted to the EGAP registry. The pre-analysis plan (PAP) associated with this project was registered in November 2018, before we had access to most of the experimental data.<sup>7</sup>

We had access to the majority of the data after substantial delays in December 2019. We registered an update to the PAP in January 2020.<sup>8</sup> The updates from the first version are not relevant to this project, as they mostly pertain to the parallel study on the effects of the community policing program. We further amended the PAP, including a specification appendix specifically for this project, before we undertook any data analysis in June of 2020.<sup>9</sup> The analysis was also registered at the AEA Registry with registration code AEARCTR-0007785. We did so when considering where to submit the paper and taking into account the AEA mandatory editorial guidelines.<sup>10</sup>

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<sup>7</sup><https://osf.io/j2p5y/>

<sup>8</sup>Available at <https://osf.io/j2p5y/>.

<sup>9</sup>Available at <https://osf.io/f923e/>.

<sup>10</sup>From January 2018, the submission policy to AEA Journals makes mandatory registration in

**Hypotheses.** In the November 2018 PAP, we registered the hypotheses to be tested that we reproduce here in Appendix Table A11. Our understanding at that point was that we would be able to distinguish which officer took specific actions within the incident. For example, we believed we would obtain data regarding which officer in the incident was responsible for use of force and who conducted arrests. We postulated the hypotheses based on this understanding. We later learned that it was not possible to distinguish which officer in an incident had been responsible for each outcome. Our outcome data is instead at the level of the *incident*, not the officer within the incident. Following on the example above, we observe if there was use of force or arrests at the level of the incident, but not the specific officer who undertook those actions. This made it impossible to test H1-a and -b, H2-a and -b, H3-a and -b, and H4-a and -b. Moreover, the low quality of the citizen complaints dataset made it impossible to test H2 altogether. Our definition of the treatment follows the hypotheses H1-c and H3-c and extrapolates the same definition to be able to test H4.

**Outcomes.** In our PAP registered on June, 2020, we describe our main outcomes as being: physical force deployed (FORCA FISICA), number of non-lethal weapons (QTD ARMA NAO LETAL), number of lethal weapons (ARMA LETAL), number of fireshots (ARMA LETAL DISPAROS), number of handcuffed (TOTAL ALGEMADOS), number of arrests (TOTALPRISOES), and number of victims (TOTAL VITIMAS). These were selected based on the variables we knew we would have access to based on conversation with the Military Police.

Given the low frequency of any use of force (including non-lethal), we opt to combine all of the outcomes on this category into groups to increase power. Physical force, non-lethal or lethal weapon deployment<sup>11</sup> are combined and included as use of force, which results are reported in column (1) of Table 2. Number of handcuffed individuals and arrests were combined and converted into a binary variable, with results reported on column (2). We combine these two main groupings with other information stemming from police reports to form our "Adverse Interaction Index" the AEA RCT registry.

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<sup>11</sup>We can only observe whether there was any gunshot fired, so in fact, gunshots fired and lethal weapon deployment are equivalent.

based on [Anderson \(2008\)](#), which is reported on column (4). This was pre-registered as we wanted to provide robustness to multiple hypothesis testing.

With respect to number of victims, we converted into a binary variable for the presence of a victim in the police report and decide to report in column (8) of [Table 2](#) as an additional margin of potential effects of body-worn cameras on reporting.

In order to follow the PAP, we estimate our main equation again, using as outcomes use of force and its components, meaning physical force, non-lethal force (including tasers, pepper spray, and others), and lethal force. These results are presented in [Appendix Table A12](#). As we can see, the reduction of 61.2% in use of force is likely driven by physical and lethal force.

The reporting outcomes (event recorded, event registered with no information, and criminal typology) are part of our own exploratory analysis. Nonetheless, given the effect sizes detected (see [Table 1](#)), we believe them to be of high importance for the understanding of the impact of body-worn cameras. The fact that the increases in reporting of certain types of crimes are larger for domestic violence and, to a lesser extent, burglary and assault are important findings, and for this reason we opted to include in the main tables.

**Specifications.** In our June, 2020 PAP, we registered the following specification as our main one:<sup>12</sup>

$$y_{ibdw} = \beta_1 \text{Treated Event}_i + \eta_{bw} + \tau_d + z_{ibdw} + \epsilon_{ibdw} \quad (8)$$

Therefore,  $i$  in the main text specification refers to the incident/event, which replaces the  $d$  in the PAP. The latter now refers to weekday. Keeping  $b$  to refer to police precinct,  $d$  and  $w$  now explicitly refers to the granular time dimension present in our study (weekdays and experimental weeks), which is important to account for due the randomization of blackout days.

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<sup>12</sup>Note that the sub-indices notation have been changed to follow the manuscript. We add the subscripts  $d$  and  $w$  to differentiate weekdays and experimental weeks from a more general  $t$  time subscript adopted in the PAP, where we nonetheless make explicit what our fixed-effects refer to.

$\tau_d$  is equivalent to  $\eta_t$  in the PAP (weekday fixed effects),  $\eta_{bw}$  is equivalent to  $\eta_{bt}$  (precinct-week fixed effects) and  $z_{ibdw}$  is equivalent to  $\eta_d$  (number of officers in the incident). We have indeed not included stratification bins fixed-effects in the pre-registered equation – this is only included in PAP’s equation (7), as a potential specification we would estimate at the officer-level. This is due to our own fault as we did not consider during registration that the stratification at the officer level would need to be translated into stratification bins at the event-level. However, we deemed this necessary when running our analysis to make sure we accounted for the stratified nature of our randomization. Reassuringly, running the specification (8) above as registered in the PAP produces similar results to the ones reported in the main text. We can provide these results upon request.

**Heterogeneities.** The event risk pre-assessment by the police was also pre-registered in the heterogeneity section of the PAP. In line with the pre-registration, explore the effects of treatment intensity in Panel C of Tables 1 and 2, as well as for the escalation potential of incidents based on *ex-ante* risk level and historical use of force on Panels D and E of Table 2. Our heterogeneity by the characteristic of the officer carrying the camera was first registered as one to be tested using officer-level data, but, given the nature of the data discussed above, we map the rank of officer into the incident characteristic to be explored. Table A3 present a summary of officer characteristics used for stratification and Table A2 on how they translate into incident level ones.

Other specifications registered in the PAP pertain to the event-study design and variations of the main equation using less granular variation aggregated at the officer and precinct levels. We use those specifications in Section 4 and Appendix Section B, as they allow us to compare the results with previous studies in the literature which implemented similar designs. Once more, we replicated all the analysis with the exact versions in the PAP and found results that are very consistent with those reported in the paper, and which are available upon request.

## D Marginal Value of Public Funds

We present the framework used to calculate the marginal value of public funds (MVPF) associated with implementing body-worn cameras (BWCs) in Santa Catarina. This calculation is based on

the causal estimates derived from our experiment. Following the approach outlined by [Finkelstein and Hendren \(2020\)](#) and [Hendren and Sprung-Keyser \(2020\)](#), we understand the MVPF as the monetary value of the societal benefits generated for every dollar spent on implementing these devices.

As we elaborate below, valuing the benefits of implementing cameras is inherently challenging, as it is not always straightforward how to monetize the benefits along many potential margins, such as the reduced use of force, increased reporting (particularly in cases of domestic violence), and the resulting rise in the number of victims and cases brought under investigation. Furthermore, this is a context where it is difficult to measure fiscal externalities.

To advance our analysis, we begin by focusing on a specific and objective outcome: the reduction in police *lethality* reasonably inferred from our experiment. From a welfare perspective, this is the most relevant benefit, since the value of losing a life represents an extraordinary cost to society. To quantify this cost, we draw on five estimates from the existing literature on the value of statistical life (VSL) that are relevant to our context. These estimates provide a broad range of VSL values, and we adopt the median value for our calculations, while also testing the robustness of our findings using the lower end of the VSL range.<sup>13</sup> As we detail below, in our preferred estimate, the benefits of the reduction in lethality alone amounts to USD 842,772.12. We express all values in 2018 US Dollars, and value both the benefits and costs per 100k inhabitants per year of intervention.

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<sup>13</sup>More specifically, we obtained five different estimates of the Value of Statistical Life to Brazil, and convert them to 2018 dollars. In increasing order, those are the studies of [Corbi et al. \(2006\)](#) and [Rocha and Soares \(2015\)](#), with a minimum VSL of USD 292,608.55; [Cerqueira \(2014\)](#) and [Cerqueira et al. \(2020\)](#) at USD 540,200.40; [Pereira et al. \(2020\)](#) at USD 816,743.99; [Viscusi and Masterman \(2017\)](#) at USD 1,685,000; and the maximum estimate of [Corbi et al. \(2006\)](#) and [Rocha and Soares \(2015\)](#), with a VSL of USD 2,250,835. We take the median value of this wide range of estimates, of [Pereira et al. \(2020\)](#) at USD 816,743.99. The mean would provide a higher figure of USD 1,137,297.59.

We then consider the benefit of a reduction in non-lethal use of force. This part of the analysis is more uncertain, as quantifying the benefits of reduced police actions such as use of pepper sprays, or tasers, is more difficult to quantify. We take estimates from the literature ([Cohen and Piquero, 2009](#), [Williams et al., 2021](#)) and re-scale them to our context. The reduction in non-lethal force yield an additional benefit of USD 186,188.89. The final piece in our analysis – the reduction of arrests and handcuffs – produce another modest gain of USD 6,620.15. Thus, taken together, this suggest a total benefit of USD 1,035,581.17.

We then look at the costing side. Putting all direct costs (cameras, docking stations) with human resource costs together (supervision, officer training) produce a cost estimate *in our experiment* of USD 111,816. Our preferred estimate of Marginal Value of Public Funds is then at 9.26.

Our estimates align with recent evidence on BWC cost-effectiveness from the United States ([Williams et al., 2021](#)) which point to benefit-cost ratio of 4.95 in that context. The particularly strong welfare gains in our setting suggest that BWCs may be especially valuable in middle-income countries where baseline levels of police violence tend to be higher and implementation costs lower. Higher MVPFs in low- or middle-income countries is also seen in other types of interventions ([Hahn et al., 2024](#)). This implies that prioritizing BWC adoption in such contexts could generate especially high returns on investment.

While the quantification above suggests a relatively high MVPF, it is important to emphasize that estimating the benefits of BWCs in this context is inherently challenging, and the MVPF presented here should be regarded as an approximation at best. As noted earlier, the estimates of the VSL are subject to significant uncertainty, with studies reporting a wide range of values. Reassuringly, even the lowest VSL estimate in our sample (the minimum values reported by [Corbi et al., 2006](#) and [Rocha and Soares, 2015](#)) results in an MVPF of 4.42, which remains well above 1. Furthermore, as we outline below, we have opted not to include certain second-order effects, such as the impact on reporting or *depolicing*. This decision reflects either the fact that their exclusion likely provides a conservative lower bound for the MVPF or that we have provided evidence suggesting null effects in these areas.

We now discuss the quantification of costs and benefits in more detail.

**Costs.** Numbers are presented in Appendix Table A14. Our RCT included the randomization of 73 cameras at the cost of USD 341.62 each. Cameras need to be charged (and have footage downloaded) on an after-shift basis on docking stations located at the precincts. Seven were purchased for the experiment, each costing USD 3,030.50. Thus the direct equipment costs are USD 46,151.76. We conservatively assume a full depreciation after one year of equipment use.

Given that the operation of BWC software and video oversight is labor-intensive, we account for the allocation of one junior officer (Capitao) and one senior officer (Coronel) to the program. This assumption is based on discussions with the Military Police and is likely an overestimate. In our experiment, the senior officer was not fully dedicated to the BWC intervention and held a lower rank (Tenente Coronel). Nonetheless, we use this estimate to account for additional minor and dispersed costs that are difficult to quantify, such as internal consultations with legal departments or similar activities. We collect their average yearly wages from publicly available data and convert it into 2018 USD.<sup>14</sup> Junior and senior police officers earn USD 90,426.40 and USD 127,881.72 yearly, respectively. We further account for police officer training: we assume that officers will spend 2 out of 200 working days per year in activities related to management of the cameras, so 1% of their time. Adding across 150 officers, we then account for an additional personnel cost of 150% of a junior officer.

This brings us to a total cost of USD 400,780.93 per year of intervention, for 150 treated officers, so USD 2,671.87 per officer-year. This is within the range of BWC cost estimates in the literature (Williams et al., 2021; Kim, 2024), although slightly on the higher end. We then linearly scale the cost as if the cameras were fully implemented to the 453 officers in our experimental precincts, and divide by the number of 100k inhabitants.<sup>15</sup> Our final estimate is then a cost of USD 111,816

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<sup>14</sup>See <https://www.transparencia.sc.gov.br/>. Obtained all personnel with rank “Capitao” and “Coronel” in the Military Police of Santa Catarina. The numbers above are averages.

<sup>15</sup>This was the intention of the Santa Catarina Military Police as our study was intended to evaluate the policy that would then be rolled-out state wide.



per 100k inhabitants-year.

**Benefit – reduction in police lethality.** We now calculate the value associated with reducing the direct occurrence of individuals being killed by police officers in the line of duty. The primary challenge lies in translating our main estimates – focused on the reduction of *any* use of force – into reasonable assumptions about the reduction in police lethality. Our data contains officer measures of use of force, including lethal use of force, defined as the discharge of firearms. However, it does not specify whether there were victims of police lethality.

We first compute the baseline incidence of police lethality. In our data, we find 13.80 incidents with lethal use of force (per 100k inhabitants-year) in the pre-experimental period. We then need to convert this into a measure of police lethality. We compared the number of incidents involving the use of lethal force in Florianopolis in our dataset (172 in 2018) with the 21 deaths caused by police actions in the same year from an external source ([Fórum Brasileiro de Segurança Pública, 2018](#)).<sup>16</sup> This comparison yields an estimated risk of lethality, conditional on police use of lethal force, of  $21/172 = 12.21\%$ . This figure could underestimate the total number of deaths occurring during police incidents, as it does not account for cases where deaths occur during police actions but cannot be directly or verifiably attributed to police officers. Based on this, we estimate approximately 1.685 police-related deaths per 100k inhabitants per year (and, conversely, 12.13 cases of police lethal use of force that did not result in lethality). Those numbers are shown in Column (1), Panel A and first row of Panel B of Table A13.

We now take our treatment estimates from two different approaches.

*Approach 1: taking estimates from Table 2.* In our first, and preferred approach, we assume that the use-of-force coefficients reported in Table 2 translate into a reduction of police lethality with the same magnitude. This indicates a 61.2% reduction, resulting in 1.03 averted death per 100k inhabitants-year, as shown in Column (4) of Panel A. We then apply the estimate of the benefit of aversion in Column (2), reaching a total benefit of USD 842,772.12. We note that VSL in Brazil

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<sup>16</sup>Unfortunately, data for other municipalities was not available.

are much smaller than in the United States (Viscusi and Masterman, 2017). This will, in turn, depress our computed benefits from reduction of lethality compared to lives in the United States.

*Approach 2: taking estimates from Table A12.* To achieve this, we focus on an indicator for lethal use of force and re-estimate the main specification from Equation (1). The complete results of this auxiliary regression are presented in Table A12. These estimates are also reproduced in Columns 6-8 of Table A13, expressed as percentage reductions relative to the mean of the dependent variable. The point estimate is approximately 120%, though capped at 100%. While the results are imprecisely estimate – hence our preference for the former approach – we still consider this to provide supporting evidence. We then assume that the observed reduction in lethal use of force corresponds directly to a reduction in police lethality. Specifically, applying these treatment estimates suggests a full aversion of police lethality, equivalent to 1.685 lives saved per 100,000 inhabitants per year. The benefits of this reduction are then valued at USD 1,376,197.30.

**Benefit – reduction in police use of force beyond lethality.** Monetizing the averted use of police force beyond cases of lethality is more nuanced, as it encompasses a wide spectrum of actions and outcomes of police action. These range from mere officer presence at the scene to the use of measures such as pepper spray, tasers, non-lethal firearm discharges, or even lethal weapon discharges (such as firearms) that do not result in fatalities. To address this, we decompose the effects based on the specific consequences of each type of force used. In every case, we reproduce the two approaches used for inferring the benefits of reduced lethality (based on the main coefficients of Table 2 and the auxiliary regression Table A12).

*Lethal weapons without lethal consequences.* In line with previous analysis, we estimate that 12.13 incidents per 100k inhabitants-year with discharge of firearms and do not result in fatalities. In Columns 3-5 of Panel B of Table A13, we apply the estimated reduction of 61.2%, and we find that 7.41 non-lethal incidents are averted. In Columns 6-8, we apply the coefficients from the auxiliary regression Table A12. Again, the former approach, with remains our preferred one, delivers more conservative estimates.

Turning this estimate into monetary terms is inherently challenging. The literature on body-worn cameras (BWC) has often relied on willingness-to-pay (WTP) measures to estimate the benefits of averting crimes and extrapolated these findings to the effects of reducing police use of force. For example, (Williams et al., 2021) quantifies the benefit of avoiding police non-lethal use of force by equating it to the cost of being a victim of aggravated assault, estimating this value at approximately USD 102,850 per incident. They use the estimates for the WTP per averted aggravated assault from Cohen and Piquero (2009).

The authors acknowledge the possibility that society perceives harm from interpersonal violence differently than harm resulting from police use of force, a perspective we share. An additional challenge is the lack of a Brazil-specific estimate for this value, unlike the availability of country-specific VSL estimates. To adjust for this, we apply a scaling factor derived by comparing the WTP of averted murder from Cohen and Piquero (2009) (USD 14,278,000) to the Brazil-specific VSL as used above (USD 816,743.99). This results in a reduction factor of  $816,743.99/14,278,000 = 0.057$ , which is likely small, and driven by the wide gap in cross-country VSL valuations. Nevertheless, it provides a likely lower bound for the consequences of non-lethal use of force in Brazil. Using this factor, we downscale the USD 102,850 value for non-lethal use of force to 5,883.33 per incident in our context.

Applying those costs, our preferred approach delivers an additional benefit of USD 43,703.91 (column 5); compared to USD 71,365.91 for the alternative estimates (column 8).

*Non-lethal weapons.* In our data, we observe 28.8 incidents of non-lethal weapon discharges per 100,000 inhabitants per year. Using the same treatment effect estimates and methodologies, we find point estimates for the benefit at USD 103,763.55 (column 5, estimates from Table 2) and USD 72,461.38 (column 8, estimates from Table A12).

*Physical force.* Physical force in the literature (Cohen and Piquero, 2009) is compared to the WTP of averting a simple assault. We again downscale those estimates to obtain a measure of USD 1,315.00 for physical force costs applicable for Brazil. In our data, we find 48.08 incidents

of physical force. Following the same process, we find that the reduction in physical force provides benefits of USD 38,721.44 (column 5, estimates from Table 2) or USD 56,417.72 (column 8, estimates from Table A12).

*Further considerations.* It is important to consider how the use-of-force effect is shaped by whether the averted use of force adheres to police procedures or constitutes abusive or excessive behavior. In instances of abusive or excessive force, the associated welfare losses are likely to be more significant, as such actions undermine the role of the police as an institution and are more likely to erode civilian trust. However, we have opted to exclude this channel from our analysis for two key reasons: first, the difficulty in assigning a monetary value to this effect, and second, that, by omitting this channel, we likely provide a lower bound for the Marginal Value of Public Funds (MVPF).

On the opposite direction, reducing the use of force could, in principle, decrease the police's capacity or willingness to restore law and order. We will not consider this channel, as our results and others in the literature generally conclude that BWCs do not decrease police effort or increase crime (Lum et al., 2019). We also have not found any effects on police effort or allocation in space (Table 3).

**Other benefits beyond use of force.** We then move into considering other benefits not related to use of force.

*Arrests and handcuffs.* The rescaled estimates for arrests/handcuffs have been valued at USD 200.73, which is the median criminal justice costs across different types of crime obtained from Cohen and Piquero (2009). With point estimates at 5.9% reduction, and a baseline estimate of 559.34 incidents, we find that this provides an additional benefit of USD 6,620.15. Coefficients are shown in Panel C.

*Reporting and victims.* We were unable to assign a monetary value to the observed increase in reporting and victim identification, particularly in cases of domestic violence. Quantifying this effect would require additional information regarding, on the one hand, the benefits to individuals

of being formally recognized as victims of crime, along with the downstream impacts on the criminal justice system, such as potential deterrence effects. It would also require consideration of the associated costs, including increased investigative workloads, judicial expenses, and the potential costs of incarceration if increased reporting leads to more convictions. While these effects are plausible, we were unable to quantify them with sufficient reliability. As a result, we have opted not to include them in our analysis.

*Complaints about police action.* A final note refers to the complaints about police action. In past studies (see [Lum et al., 2019](#)), the primary outcome has been the reduction of civilian complaints against the police. Consistent with this, past cost-benefit analyses (see [Williams et al., 2021](#)) base their benefits on the averted time and investigative effort, and still find positive cost-benefit ratios in favor of BWCs of around 4.95. We, unfortunately, cannot include this in the analysis as the number of complaints against the police was not made available to us.

**Summary.** Panel D adds across all components, and shows a point estimate of the MVPF at 9.26 from our preferred approach (and 14.06 using the alternative estimates). Our MVPF is still larger than one when we consider the lowest values in the range of VSL estimates. Our preferred MVPF is comparable in terms of returns to society to early childhood programs in the US, such as Abecedarian ([Hendren and Sprung-Keyser, 2020](#)). [Deshpande and Mueller-Smith \(2022\)](#) finds similar range MVPF of Social Security Income losses on youth criminal charges, when accounting for the victim costs. We also note that our estimates are likely to be a lower bound, because we used relatively conservative estimates throughout and omitted channels that would be likely to provide higher valuations, such as the increase in reporting and absence of effects on complaints against the police action.

**Comparing to other interventions and why our estimates are high?** Our MVPF estimates are relatively high compared to those of BWC interventions ([Williams et al., 2021](#)) and other interventions with the police ([Chalfin and McCrary, 2018](#), [Facchetti, 2021](#), [Dube et al., 2023](#)). These higher estimates result from a combination of factors that highlight why BWCs may be

especially impactful in Latin America, and Brazil in particular. While costs can be relatively similar, in our context, the baseline levels of police lethality are much higher than in the United States (See Figure A2) which, combined with a large estimate of reduction, results in a large benefit.

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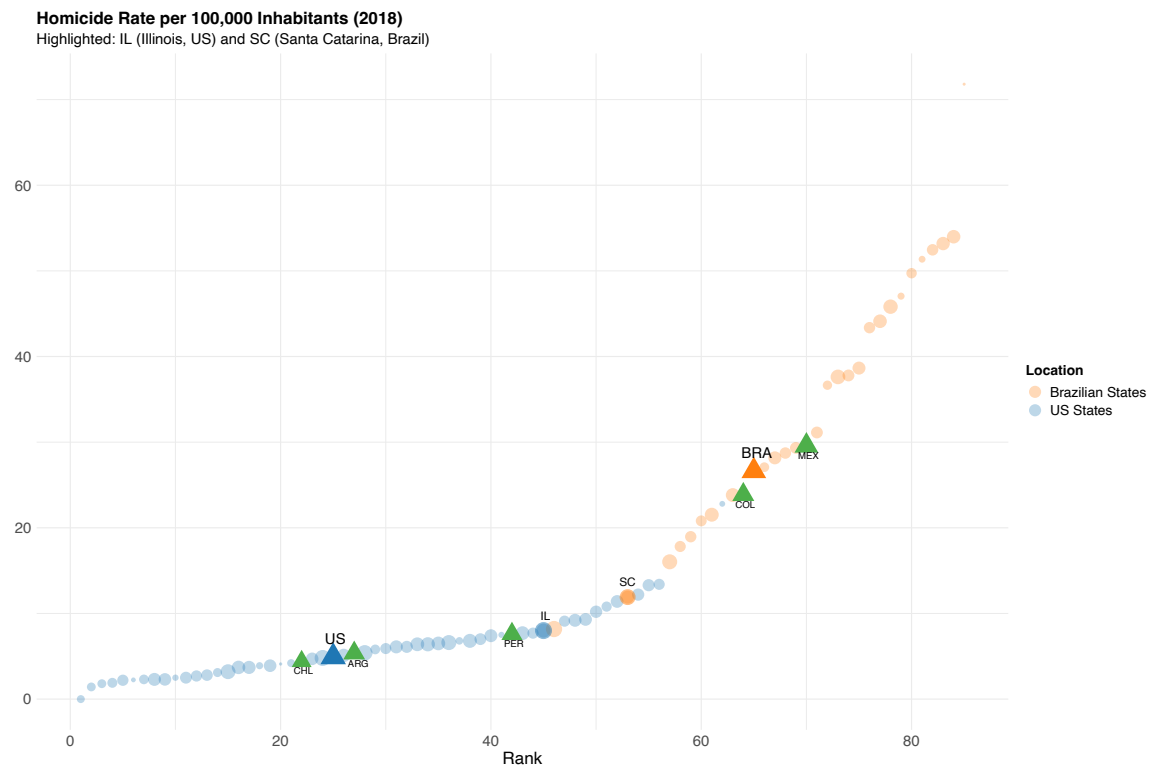
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Çubukçu, S., N. M. Sahin, E. Tekin, and V. Topalli (2021). Body-worn cameras and adjudication of citizen complaints of police misconduct. *NBER Working Paper No. 29019 July*.

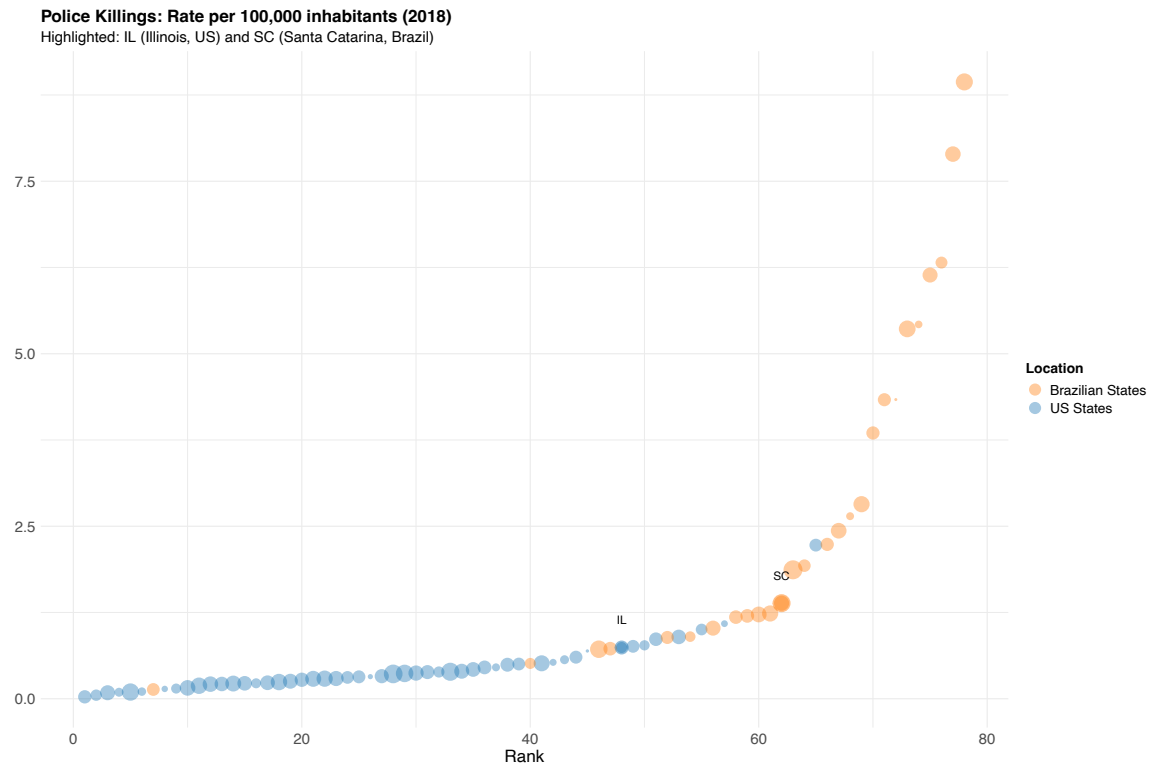
## Appendix Tables and Figures

**Figure A1:** Homicide Rate per 100,000 inhabitants



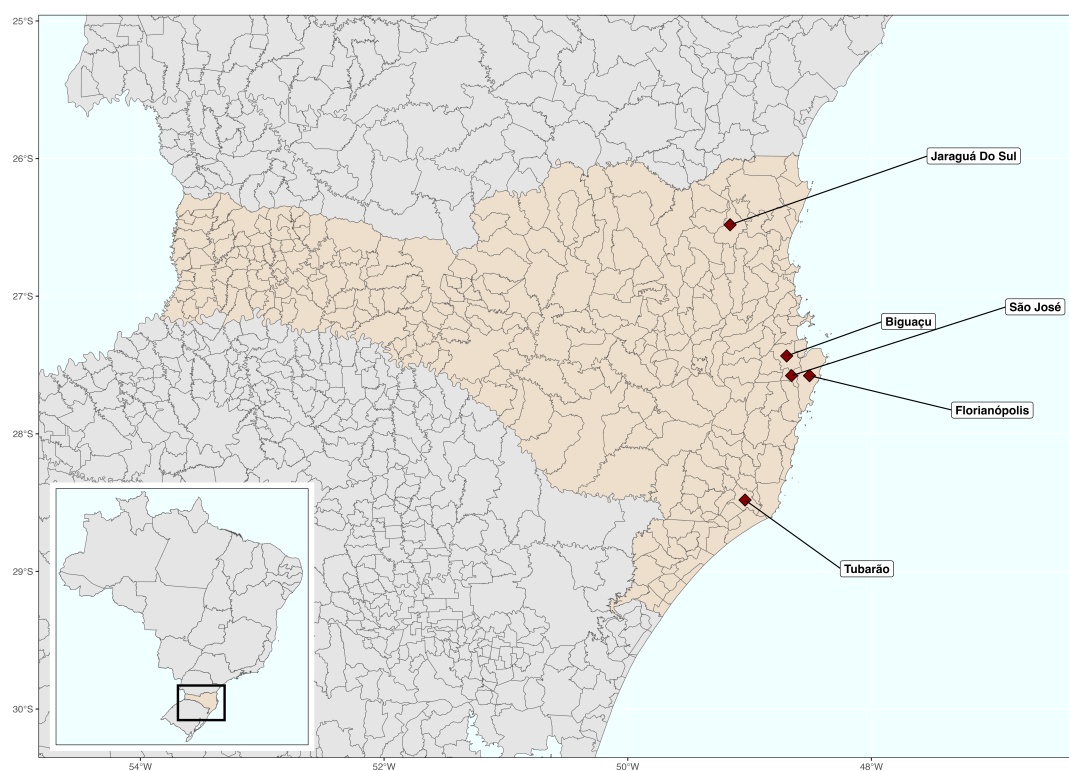
*Notes:* Source: Country level data comes from UNODC, data for the Brazilian states comes from *Atlas da Violência - Ipea*, and data for the US states is from the CDC (National Center of Health Statistics) for all states but the District of Columbia, where we get the data from the FBI. The y-axis represents the rank in terms of homicide rate, and the homicide rate is weighted by the log of the population in each location, state or country.

**Figure A2: Police Killings Rate - Comparison Brazil and US**



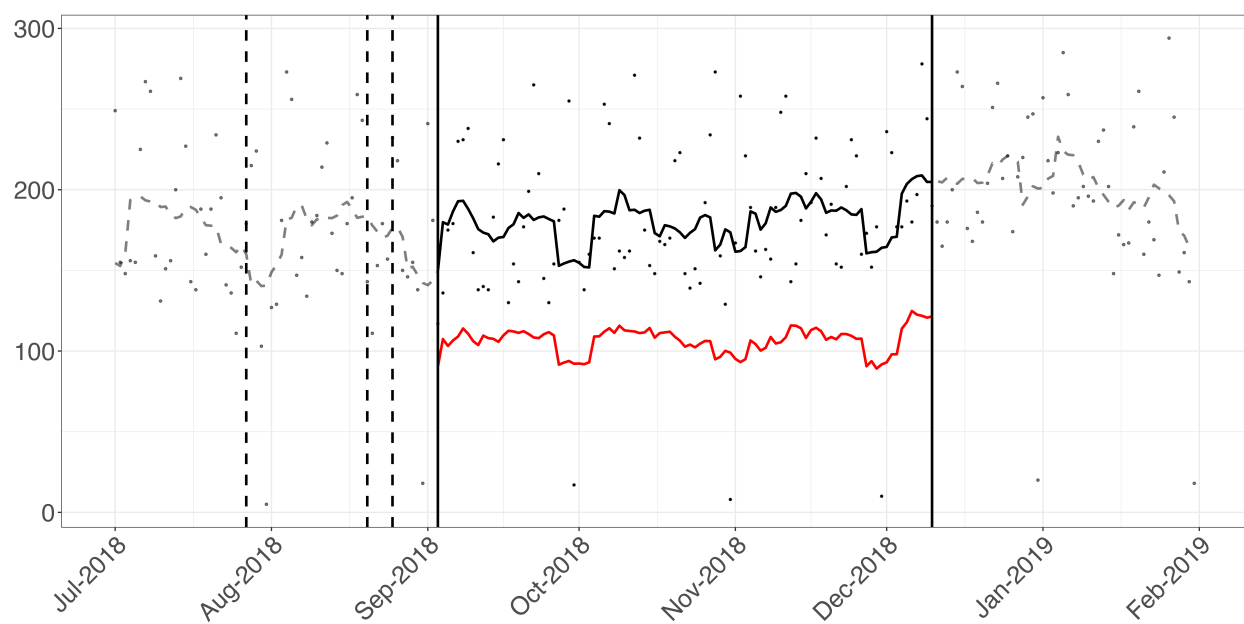
*Notes:* For each state, we calculate the rate of police killings per 100,000 inhabitants with data from the US and Brazil. US police killings data from Mapping Police Violence project, and BR police killings data is from the *Anuario Brasileiro de Segurança Pública*, a yearly dataset that compiles and reports data about public safety, crime and policing. The y-axis represents the rank in terms of police killings rate, and the data is weighted by the log of the population in each location, state or country.

**Figure A3:** State of Santa Catarina and the experimental sites where the BWC intervention was implemented



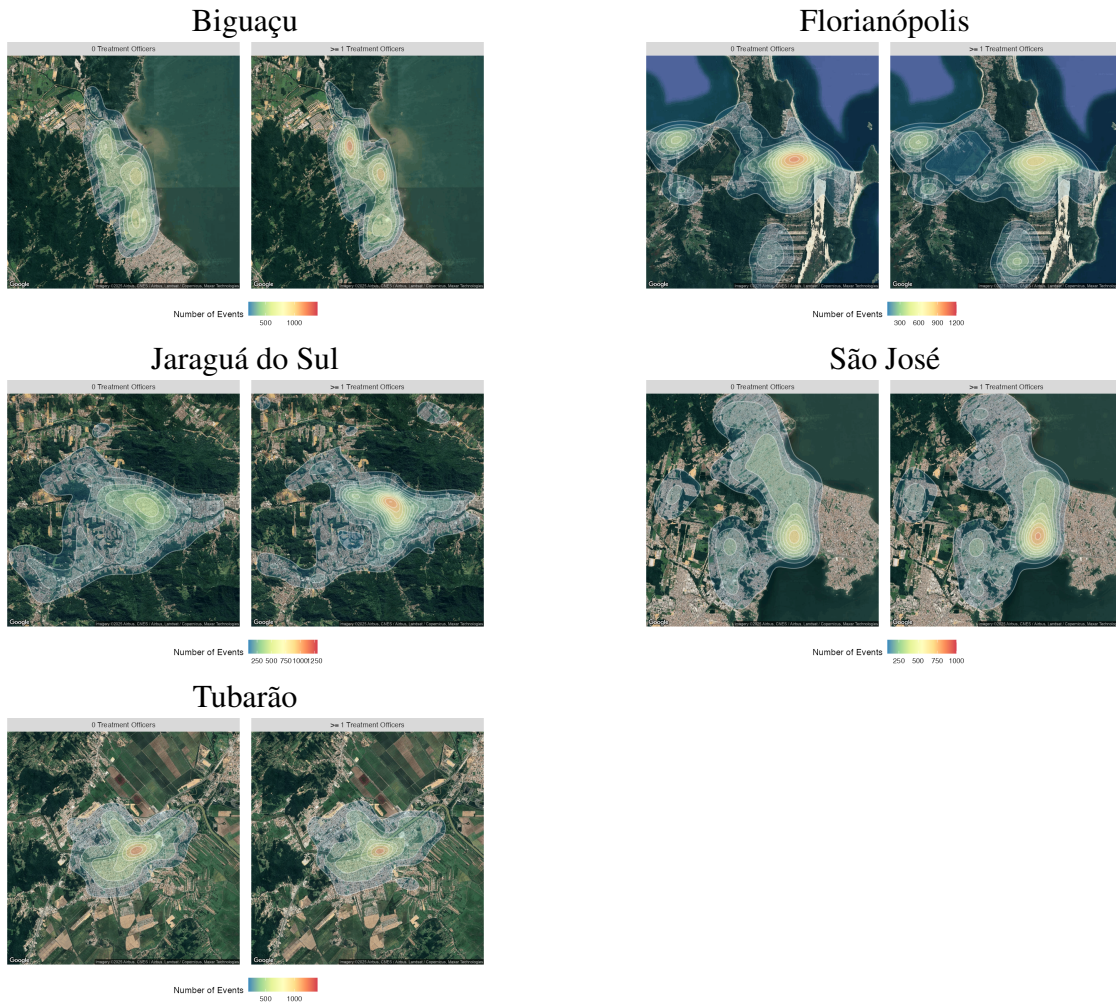
*Notes:* This figure presents the experimental sites where the experiment took place. These are the catchment areas of the 24<sup>th</sup> Police Precinct in Biguaçu, the 21<sup>st</sup> in Florianópolis, the 14<sup>th</sup> in Jaraguá do Sul, the 7<sup>th</sup> in São José, and the 5<sup>th</sup> in Tubarão.

**Figure A4:** Time series of treated and control events



*Notes:* This figure presents the time series of the events by treatment status. The solid lines measure the seven-day moving average of the number of treated events (in red) and the overall number of events, illustrating that, on average, 50% of events had the presence of a BWC. Each dot measures the number of events in a given day. The dashed vertical lines represent important dates of the experiment design, as illustrated in Figure 1.

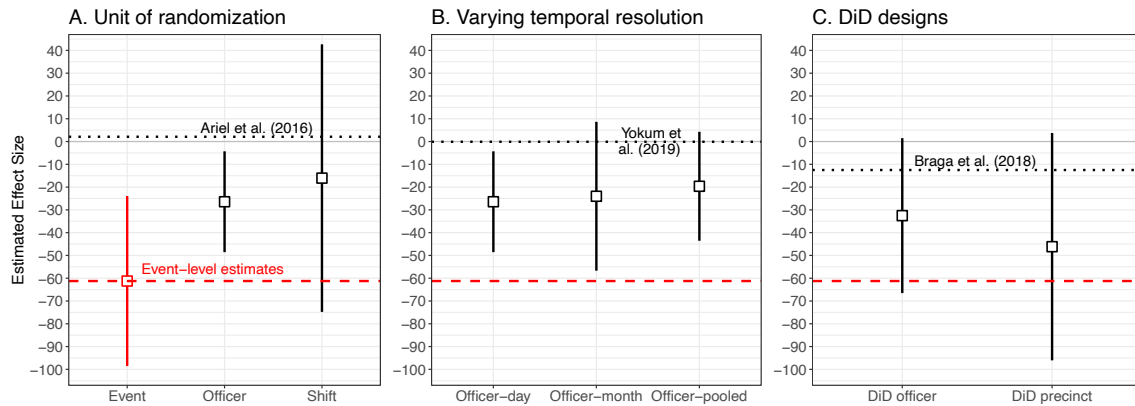
**Figure A5:** Spatial distribution of treatment and control incidents



*Notes:* This figure presents kernel density estimates of the spatial distribution of the treatment and control events across the five cities that were part of the experiment. It highlights that the spatial distribution of both the treatment and control incidents were very similar throughout and highlights the different topographies of the study area.

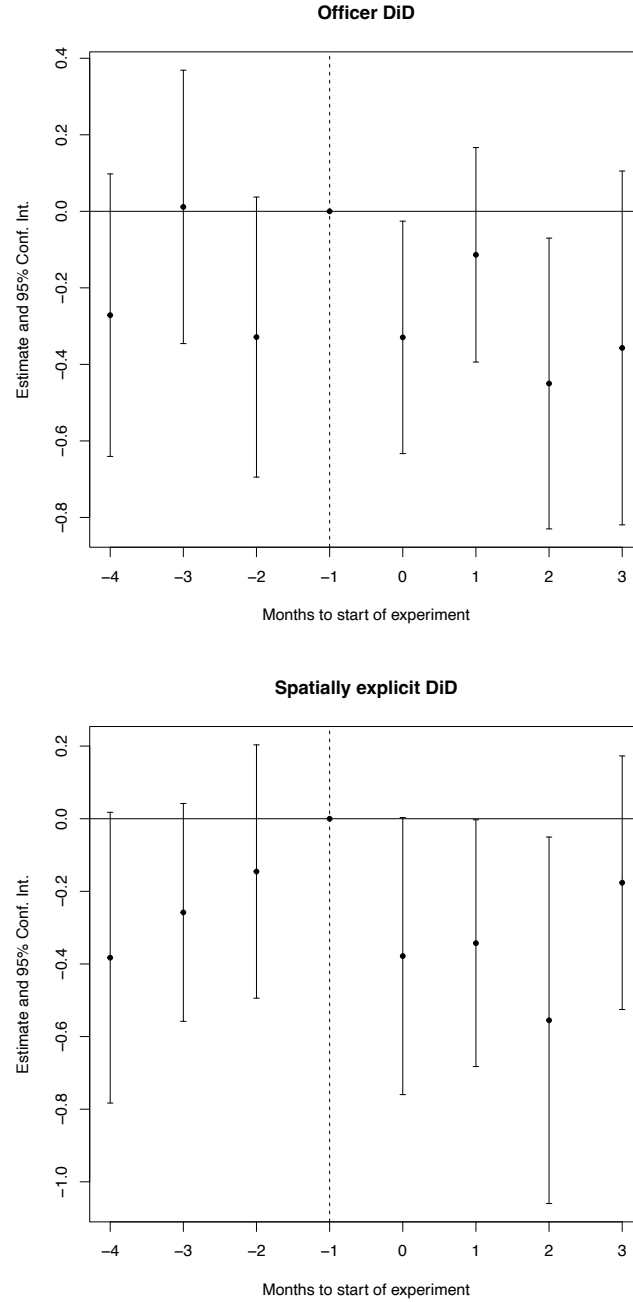


**Figure A6: Comparing the distribution of effects with different designs and the literature**



*Notes:* This figure presents the results on how the estimated treatment-effect sizes vary if we reanalyze the data using different commonly used evaluation strategies. Benchmark results from this paper exploit event-level variation and are presented in red. Estimates of effect sizes from reference studies in the literature using such designs are annotated as a horizontal dashed line and are closer to the most comparable estimate from our data. Panel A explores how changing the unit of randomization affects the results, exploring experimental variation between treated and control officers and between treated and control shifts. Panel B explores varying the temporal resolution in the aggregation of the outcome data while keeping the experimental variation of officers constant. Panel C explores two difference-in-differences models, the first exploring experimental variation between officers and the second exploring the spatially explicit implementation of BWCs.

**Figure A7: The effects of BWCs on use of force: two event-studies**



*Notes:* These figures show event-study estimates of the effects of BWCs. The first explores the variation between treated and control officers and the second the variation between treated and control precincts. The point estimates are the coefficient of the treatment unit interacted with period dummies. The officer-level DiD regression uses officer, precinct, week and weekday fixed effects and the standard errors are clustered at the precinct-week level. The spatially explicit DiD uses precinct, week, and weekday fixed effects and the standard errors are clustered at the precinct-week level.

**Table A1:** Summary statistics of study sites

|                                                     | Biguaçu  | Florianópolis | Jaraguá Do Sul | São José | Tubarão  | SC average           |
|-----------------------------------------------------|----------|---------------|----------------|----------|----------|----------------------|
| <b>Panel A. Socioeconomic Characteristics</b>       |          |               |                |          |          |                      |
| Population                                          | 58,206   | 421,240       | 143,123        | 209,804  | 97,235   | 18,468<br>(42,990)   |
| Urban (%)                                           | 0.904    | 0.964         | 0.932          | 0.989    | 0.907    | 0.599<br>(0.231)     |
| Income                                              | 1,208.22 | 2,578.28      | 1,586.99       | 1,692.74 | 1566.36  | 1,127.35<br>(236.72) |
| White (%)                                           | 0.836    | 0.846         | 0.864          | 0.844    | 0.908    | 0.829<br>(0.103)     |
| Primary school or less (%)                          | 0.292    | 0.623         | 0.594          | 0.574    | 0.656    | 0.571<br>(0.082)     |
| High school or less (%)                             | 0.797    | 0.959         | 0.941          | 0.937    | 0.965    | 0.940<br>(0.025)     |
| Water access (%)                                    | 0.995    | 0.999         | 0.996          | 0.999    | 0.995    | 0.987<br>(0.023)     |
| Computer (%)                                        | 0.490    | 0.727         | 0.585          | 0.661    | 0.569    | 0.365<br>(0.110)     |
| Internet (%)                                        | 0.391    | 0.650         | 0.427          | 0.564    | 0.462    | 0.248<br>(0.101)     |
| <b>Panel B. Violence and Use-of-Force Incidence</b> |          |               |                |          |          |                      |
| Use-of-Force Incidents                              | 23       | 52            | 34             | 62       | 22       | -                    |
| Crime Events                                        | 739      | 2135          | 2622           | 3097     | 1309     | -                    |
| Homicide Rate per 100k                              | 22.9     | 17.16         | 5.38           | 16.9     | 9.65     | -                    |
| Use of Force - Yearly Rate per 100k                 | 106.90   | 33.39         | 64.27          | 79.95    | 61.21    | -                    |
| Crime Events - Yearly Rate per 100k                 | 3,435.05 | 1,371.276     | 4,956.55       | 3,993.78 | 3,642.28 | -                    |

*Notes:* This table presents socio-demographic characteristics and baseline violence across the five study sites and the average in Santa Catarina State. The sociodemographic data comes from 2010 IBGE Census, the homicide rate from the 2016 IPEA Atlas da Violência, and use-of-force and crime-events incidence from the authors' calculations using PMSC data from March to July 14, 2018. Income is in Brazilian reais per month. Standard errors are in parentheses.

**Table A2: Event Characteristics Summary****Panel A. Officer Rank**

| # Events<br>Treated by<br>Soldier Rank | # Events<br>Treated by<br>Above Soldier<br>Rank | # Events<br>Treated by<br>Both Ranks | # Control<br>Events |
|----------------------------------------|-------------------------------------------------|--------------------------------------|---------------------|
| (1)                                    | (2)                                             | (3)                                  | (4)                 |
| 5900                                   | 1186                                            | 715                                  | 5473                |
| [44.4%]                                | [8.9%]                                          | [5.4%]                               | [41.2%]             |

**Panel B. Treatment Intensity**

| # Events<br>Treated by 1<br>Camera | # Events<br>Treated by 2 or<br>More Cameras | # Control<br>Events |
|------------------------------------|---------------------------------------------|---------------------|
| (1)                                | (2)                                         | (3)                 |
| 5787                               | 2014                                        | 5473                |
| [43.6%]                            | [15.2%]                                     | [41.2%]             |

**Panel C. Ex-Ante Event Risk Assessment**

| # Treated<br>High Risk<br>Events | # Control High<br>Risk Events | # Treated<br>Low Risk<br>Events | # Control<br>Low Risk<br>Events |
|----------------------------------|-------------------------------|---------------------------------|---------------------------------|
| (1)                              | (2)                           | (3)                             | (4)                             |
| 910                              | 568                           | 6891                            | 4905                            |
| [6.9%]                           | [4.3%]                        | [51.9%]                         | [37.0%]                         |

**Panel D. Baseline Use of Force (UoF)**

| # Treated<br>Above<br>Median UoF<br>Events | # Control<br>Above Median<br>UoF Events | # Treated<br>Below<br>Median UoF<br>Events | # Control<br>Below<br>Median UoF<br>Events |
|--------------------------------------------|-----------------------------------------|--------------------------------------------|--------------------------------------------|
| (1)                                        | (2)                                     | (3)                                        | (4)                                        |
| 1603                                       | 1099                                    | 6198                                       | 4374                                       |
| [12.1%]                                    | [8.3%]                                  | [46.7%]                                    | [33.0%]                                    |

*Notes:* Panel A reports the breakdown of total events in the experimental periods by the heterogeneities investigated in Tables 1 and 2. For all panels, the shares reported in square brackets reflect the share of events with these characteristics out of the total events (hence, rows add to 100%).

**Table A3: Officer Characteristics Summary**

|                                                   | Number of<br>Officers | Treated<br>Officers | Share<br>Treated |
|---------------------------------------------------|-----------------------|---------------------|------------------|
|                                                   | (1)                   | (2)                 | (3)              |
| <b>Panel A. Types of Activity</b>                 |                       |                     |                  |
| Preventive                                        | 19                    | 8                   | 42.1             |
| Tactical                                          | 128                   | 41                  | 32.0             |
| Regular                                           | 298                   | 100                 | 33.6             |
| <b>Panel B. Rank</b>                              |                       |                     |                  |
| Soldier Rank                                      | 297                   | 97                  | 32.7             |
| Above Soldier Rank                                | 148                   | 52                  | 35.1             |
| <b>Panel C. Previous Internal Investigations</b>  |                       |                     |                  |
| Below median IPM/PAD/SIND                         | 231                   | 78                  | 33.8             |
| Above median IPM/PAD/SIND                         | 214                   | 71                  | 33.2             |
| <b>Panel D. Gender</b>                            |                       |                     |                  |
| Female                                            | 29                    | 14                  | 48.3             |
| Male                                              | 416                   | 135                 | 32.5             |
| <b>Panel E. Precinct</b>                          |                       |                     |                  |
| Biguaçu                                           | 39                    | 12                  | 30.8             |
| Tubarão                                           | 75                    | 26                  | 34.7             |
| Jaraguá do Sul                                    | 78                    | 26                  | 33.3             |
| Florianópolis                                     | 108                   | 36                  | 33.3             |
| São José                                          | 145                   | 49                  | 33.8             |
| <b>Summary # Officers per Stratification Bins</b> |                       |                     |                  |
| Mean                                              | 11.4                  |                     |                  |
| Median                                            | 9                     |                     |                  |
| Min                                               | 2                     |                     |                  |
| Max                                               | 39                    |                     |                  |
| Total                                             | 445                   |                     |                  |

*Notes:* Table presents summary statistics of officer characteristics that were used stratify the allocation of cameras. Columns present the total number of officers, the number of officers that were randomly allocated to carry cameras, and the share of treated officers, showing the correct implementation of the one-third randomization. Panel A provides the number by the types of activity the officers perform, Panel B the number of officer above and below the median rank in the experiment, Panel C provides the breakdown by the median of the number of police complaints filled in the pre-period (which unfortunately we don't have access besides the level at the point of randomization), Panel D provides the numbers by gender and Panel E the total number of officers by precinct. We finally provide some summary statistics for the number of officers by stratification bins.

**Table A4: Effects of BWCs on reporting — blackout specifications**

|                               | Event<br>Recorded                | Event<br>Registered<br>with No Info | Verbal<br>Attrition/<br>Threat   | Noise<br>Complaint            | Burglary                       | Assault                       | Domestic<br>Violence          | Share of<br>Reports with<br>Victims | Generated<br>Investigative<br>Report |
|-------------------------------|----------------------------------|-------------------------------------|----------------------------------|-------------------------------|--------------------------------|-------------------------------|-------------------------------|-------------------------------------|--------------------------------------|
|                               | (1)                              | (2)                                 | (3)                              | (4)                           | (5)                            | (6)                           | (7)                           | (8)                                 | (9)                                  |
| Treated Event on Blackout Day | 3.343***<br>(0.785)<br>p = 0.272 | -6.631***<br>(1.802)<br>p = 0.000   | 3.193***<br>(0.971)<br>p = 0.000 | 0.178<br>(0.795)<br>p = 0.813 | -0.940<br>(0.845)<br>p = 0.088 | 0.664<br>(0.606)<br>p = 0.121 | 0.254<br>(0.513)<br>p = 0.499 | 0.435<br>(1.220)<br>p = 0.659       | 3.233*<br>(1.908)<br>p = 0.049       |
| Mean Dep. Var.                | 0.000                            | 48.937                              | 7.237                            | 8.329                         | 6.433                          | 3.848                         | 1.895                         | 17.231                              | 34.406                               |
| N                             | 4391                             | 4391                                | 4391                             | 4391                          | 4391                           | 4391                          | 4391                          | 4391                                | 4391                                 |

*Notes:* This table documents the effects of BWCs on recording and reporting measures, including criminal typology, in control shifts (when BWCs were not handed out to officers). The dependent variables are “event recorded”, indicating that the incident was partially or fully recorded using the BWC; “event registered with no info” indicating no criminal typology was recorded; the five most frequent criminal typologies reported: “verbal attrition/threat”, “noise complaint”, “burglary”, “assault” and “domestic violence”; and “share of reports with victims” and “generated investigative report”, when officers reported events to the Civil Police, who would proceed with investigations. All dependent variables are multiplied by 100. Specifications include police precinct-by-week, day of the week, number of officers and stratification bins fixed effects. The sample only includes shifts without a camera and the regression follows specification (1). Standard errors are clustered at the precinct-by-week level. \*\*\* p<0.01; \*\* p<0.05; \* p<0.1. The randomization inference p-values are indicated below the standard errors.

**Table A5:** Effects of BWCs on use of force — blackout specifications

|                               | Use of Force                  | Handcuff<br>and/or<br>Arrest  | Contempt,<br>Resistance<br>and/or Dis-<br>obedience | Adverse<br>Interaction<br>Index |
|-------------------------------|-------------------------------|-------------------------------|-----------------------------------------------------|---------------------------------|
|                               | (1)                           | (2)                           | (3)                                                 | (4)                             |
| Treated Event on Blackout Day | 0.077<br>(0.234)<br>p = 0.632 | 0.084<br>(0.809)<br>p = 0.879 | -0.001<br>(0.310)<br>p = 0.997                      | 0.051<br>(0.178)<br>p = 0.768   |
| Mean Dep. Var. Control Events | 0.517                         | 5.399                         | 0.804                                               | 0.629                           |
| N                             | 4391                          | 4391                          | 4391                                                | 4391                            |

*Notes:* This table shows the effect of BWCs on use of force in shifts when cameras were not present. The dependent variables are “use of force”, which indicates if there was any deployment of physical, non-lethal (mechanical), or lethal force by the police, not considering use of handcuff or arrest; “arrest and/or the use of handcuffs”, which is an indicator for if handcuffs were used or if any arrests made; “contempt, resist, and/or disobey”, which is an indicator for if charges of contempt, disobedience, or resistance toward the police were registered; “Adverse Interaction Index” is the standardized inverse-covariance weighted average of the three indicators in the group. Specifications include police precinct-by-week, day of the week, number of officers and stratification bins fixed effects. The sample only includes shifts without a camera. Standard errors are clustered at the precinct-by-week level. \*\*\* p<0.01; \*\* p<0.05; \* p<0.1. The randomization inference p-values are indicated below the standard errors.

**Table A6:** Testing for endogenous allocation of BWC to events — heterogeneity by officer rank

|                                                           | High<br>Baseline<br>Use of<br>Force | Latitude            | Longitude           | High<br>Baseline<br>Income | Time to<br>Dispatch<br>(Minutes) | Time to<br>Dispatch<br>Greater<br>than 5<br>min. | Active<br>Policing | Telephone<br>Initiated<br>Incidents | High<br>Ex-Ante<br>Risk |
|-----------------------------------------------------------|-------------------------------------|---------------------|---------------------|----------------------------|----------------------------------|--------------------------------------------------|--------------------|-------------------------------------|-------------------------|
|                                                           | (1)                                 | (2)                 | (3)                 | (4)                        | (5)                              | (6)                                              | (7)                | (8)                                 | (9)                     |
| Treated Event by Officer(s) with Soldier Rank             | -0.676<br>(0.969)                   | 0.000<br>(0.001)    | 0.000<br>(0.001)    | 0.821<br>(1.444)           | -1.859<br>(1.275)                | -1.615<br>(1.198)                                | -0.310<br>(0.570)  | 0.314<br>(0.573)                    | -0.292<br>(0.691)       |
|                                                           | p = 0.541                           | p = 0.892           | p = 0.913           | p = 0.792                  | p = 0.238                        | p = 0.198                                        | p = 0.744          | p = 0.745                           | p = 0.674               |
| Treated Event by Officer(s) with Higher-than-Soldier Rank | 2.176<br>(1.556)                    | 0.000<br>(0.001)    | 0.009***<br>(0.002) | 6.961***<br>(2.208)        | -1.475<br>(1.288)                | -2.000<br>(2.045)                                | 0.125<br>(1.281)   | 0.027<br>(1.299)                    | -0.112<br>(1.250)       |
|                                                           | p = 0.382                           | p = 0.907           | p = 0.068           | p = 0.211                  | p = 0.477                        | p = 0.560                                        | p = 0.960          | p = 0.993                           | p = 0.937               |
| Treated Event by Officers of Both Types                   | 0.560<br>(2.067)                    | 0.006***<br>(0.001) | 0.012***<br>(0.002) | 0.659<br>(3.144)           | -0.462<br>(3.311)                | 4.224<br>(2.580)                                 | 0.692<br>(1.847)   | -0.535<br>(1.837)                   | -1.654<br>(1.600)       |
|                                                           | p = 0.825                           | p = 0.037           | p = 0.026           | p = 0.915                  | p = 0.846                        | p = 0.180                                        | p = 0.826          | p = 0.862                           | p = 0.396               |
| Mean Dep. Var.                                            | 20.080                              | -27.468             | -48.787             | 48.639                     | 10.701                           | 43.868                                           | 7.637              | 92.180                              | 10.378                  |
| N                                                         | 13274                               | 13274               | 13274               | 13274                      | 13274                            | 13274                                            | 13274              | 13274                               | 13274                   |

*Notes:* This table presents tests for the characteristics of the incident that could suggest endogenous allocation with respect to treatment assignment. The dependent variables are defined as in Table 3. The sample includes all events in the experimental period and excluded blackout shifts. The specification includes police precinct-by-week, day of the week, number of officers and stratification bins fixed effects. Standard errors are clustered at the precinct-by-week level. \*\*\* p<0.01; \*\* p<0.05; \* p<0.1. The randomization inference p-values are indicated below the standard errors.



**Table A7:** Correlation between event characteristics and officer rank

|                                          | High Risk<br>(1) |
|------------------------------------------|------------------|
| Event with High-Ranking Officer          | 0.939<br>(1.069) |
| Mean Dep. Var. Events Only With Soldiers | 9.820            |
| <i>N</i>                                 | 13274            |

*Notes:* This table shows the correlation between the presence of a high-ranking officer in an event and the ex-ante level of risk of an event. High-risk is the ex-ante risk assessment indicator used in Table 2. The specifications include police precinct-by-week, day of the week, number of officers and stratification bins fixed effects. Standard errors are clustered at the precinct-by-week level. \*\*\*  $p < 0.01$ ; \*\*  $p < 0.05$ ; \*  $p < 0.1$ .

**Table A8: Sample Robustness**

|                                             | Reporting Margin     |                               |                     |                               |                                |                      |                        |                                           |                           |
|---------------------------------------------|----------------------|-------------------------------|---------------------|-------------------------------|--------------------------------|----------------------|------------------------|-------------------------------------------|---------------------------|
|                                             | Event Recorded       | Event Registered with No Info | Domestic Violence   | Share of Reports with Victims | Generated Investigative Report | Use of Force         | Handcuff and/or Arrest | Contempt, Resistance and/or Dis-obedience | Adverse Interaction Index |
|                                             | (1)                  | (2)                           | (3)                 | (4)                           | (5)                            | (6)                  | (7)                    | (8)                                       | (9)                       |
| Panel A. Main Results                       |                      |                               |                     |                               |                                |                      |                        |                                           |                           |
| Treated Event                               | 24,043***<br>(1,873) | -2,770**<br>(1,239)           | 1,138***<br>(0,351) | 2,783***<br>(0,805)           | 3,101**<br>(1,190)             | -0,425***<br>(0,157) | -0,320<br>(0,471)      | -0,263<br>(0,196)                         | -0,371**<br>(0,149)       |
| Mean Dep. Var.                              | 0,000                | 47,268                        | 1,644               | 13,832                        | 32,761                         | 0,694                | 5,427                  | 0,932                                     | 0,790                     |
| N                                           | 13274                | 13274                         | 13274               | 13274                         | 13274                          | 13274                | 13274                  | 13274                                     | 13274                     |
| Panel B. Including Blackout Days            |                      |                               |                     |                               |                                |                      |                        |                                           |                           |
| Treated Event                               | 18,852***<br>(1,521) | -3,647***<br>(1,046)          | 0,899***<br>(0,286) | 2,180***<br>(0,652)           | 3,100***<br>(1,097)            | -0,307**<br>(0,120)  | -0,171<br>(0,429)      | -0,190<br>(0,172)                         | -0,268**<br>(0,113)       |
| Mean Dep. Var.                              | 0,000                | 47,671                        | 1,705               | 14,652                        | 33,158                         | 0,652                | 5,420                  | 0,901                                     | 0,751                     |
| N                                           | 17665                | 17665                         | 17665               | 17665                         | 17665                          | 17665                | 17665                  | 17665                                     | 17665                     |
| Panel C. Two Officers - Modal Dispatch Size |                      |                               |                     |                               |                                |                      |                        |                                           |                           |
| Treated Event                               | 24,140***<br>(1,930) | -3,641**<br>(1,416)           | 1,102***<br>(0,336) | 3,365***<br>(0,897)           | 4,036***<br>(1,328)            | -0,222*<br>(0,114)   | -0,158<br>(0,474)      | -0,371*<br>(0,205)                        | -0,271**<br>(0,125)       |
| Mean Dep. Var.                              | 0,000                | 49,147                        | 1,575               | 13,173                        | 30,569                         | 0,416                | 3,807                  | 0,810                                     | 0,557                     |
| N                                           | 9928                 | 9928                          | 9928                | 9928                          | 9928                           | 9928                 | 9928                   | 9928                                      | 9928                      |
| Panel D. At Most Four Officers              |                      |                               |                     |                               |                                |                      |                        |                                           |                           |
| Treated Event                               | 24,061***<br>(1,859) | -3,024**<br>(1,268)           | 1,164***<br>(0,339) | 2,947***<br>(0,848)           | 3,196**<br>(1,236)             | -0,344***<br>(0,128) | -0,427<br>(0,433)      | -0,285<br>(0,177)                         | -0,325**<br>(0,125)       |
| Mean Dep. Var.                              | 0,000                | 47,572                        | 1,619               | 13,600                        | 32,335                         | 0,595                | 5,153                  | 0,856                                     | 0,698                     |
| N                                           | 12546                | 12546                         | 12546               | 12546                         | 12546                          | 12546                | 12546                  | 12546                                     | 12546                     |
| Panel E. Pre-Experimental Data              |                      |                               |                     |                               |                                |                      |                        |                                           |                           |
| Treated Event                               |                      | 0,554<br>(0,785)              | 0,250<br>(0,167)    | 0,433<br>(0,500)              | 1,184*<br>(0,703)              | -0,036<br>(0,098)    | -0,043<br>(0,280)      | -0,006<br>(0,128)                         | -0,026<br>(0,091)         |
| Mean Dep. Var.                              |                      | 44,725                        | 2,114               | 15,272                        | 33,735                         | 0,709                | 5,606                  | 1,010                                     | 0,826                     |
| N                                           |                      | 35659                         | 35659               | 35659                         | 35659                          | 35659                | 35659                  | 35659                                     | 35659                     |

*Notes:* This table presents intention-to-treat specifications. The unit of observation is a police event. The dependent variables defined as in Tables 1 and 2. The specifications include police precinct-by-week, day of the week, number of officers and stratification bins fixed effects. Shifts without cameras are excluded from the regression. Standard errors are clustered at the precinct-by-week level. \*\*\* p<0.01; \*\* p<0.05; \* p<0.1.

**Table A9: Characteristics of BWC Studies in the Literature**

| Paper                                                   | Year<br>(1) | # Citations<br>(2) | Country<br>(3) | BWC varies<br>across:<br>(5) | T<br>(6) | C<br>(7) | Analysis<br>Unit<br>(8) | Analysis<br>sample size<br>(9) | Share of<br>Treated<br>Units<br>(10) | Avg # of<br>officers per<br>incident<br>(11) | UoF as<br>outcome?<br>(12) | Effects<br>on UoF<br>(13) | Empirical<br>strategy<br>(14) |
|---------------------------------------------------------|-------------|--------------------|----------------|------------------------------|----------|----------|-------------------------|--------------------------------|--------------------------------------|----------------------------------------------|----------------------------|---------------------------|-------------------------------|
| <i>Panel A: Shift-centric studies (7 studies)</i>       |             |                    |                |                              |          |          |                         |                                |                                      |                                              |                            |                           |                               |
| Ariel et al. (2015)                                     | 2014        | 633                | US             | Shift                        | 489      | 499      | Shift                   | 988                            | 0.495                                | 1                                            | Yes                        | Null                      | Poisson regression            |
| Ariel et al. (2016b)                                    | 2016        | 287                | Multisite      | Shift                        | 2,447    | 2,468    | Shift                   | 4,915                          | 0.498                                | -                                            | Yes                        | Increase                  | Cohen's d                     |
| Ariel et al. (2016a)                                    | 2016        | 218                | Multisite      | Shift                        | 2,447    | 2,468    | Shift                   | 4,915                          | 0.498                                | -                                            | Yes                        | Mixed                     | Cohen's d                     |
| Ariel et al. (2017)                                     | 2017        | 243                | Multisite      | Shift                        | 1,908    | 1,974    | Shift                   | 3,882                          | 0.491                                | -                                            | No                         | -                         | Cohen's d                     |
| Henstock and Ariel (2017)                               | 2017        | 107                | UK             | Shift                        | 215      | 215      | Shift                   | 430                            | 0.5                                  | 1                                            | Yes                        | Decrease                  | Odds-Ratio                    |
| Ariel et al. (2018)                                     | 2018        | 86                 | Multisite      | Shift                        | 3,637    | 3,697    | Shift                   | 7,334                          | 0.495                                | -                                            | No                         | -                         | Odds-Ratio                    |
| Magaloni (2019)                                         | 2019        | 2                  | BR             | Unit-shift                   | 16,390   | 18,642   | Shift                   | 35,032                         | 0.468                                | 1+                                           | Yes                        | Null                      | OLS                           |
| <i>Panel B: Officer-centric studies (11 studies)</i>    |             |                    |                |                              |          |          |                         |                                |                                      |                                              |                            |                           |                               |
| Jennings et al. (2015)                                  | 2015        | 299                | US             | Officer                      | 46       | 43       | Officer                 | 89                             | 0.517                                | 1+                                           | Yes                        | Unclear                   | t-test                        |
| Ready and Young (2015)                                  | 2015        | 288                | US             | Officer                      | 50       | 50       | Contact report          | 3,698                          | 0.5                                  | 1+                                           | No                         | -                         | Logit                         |
| White et al. (2017)                                     | 2017        | 167                | US             | Officer                      | 82       | 67       | Officer*time            | 298                            | 0.55                                 | -                                            | Yes                        | Null                      | DiD                           |
| Jennings et al. (2017)                                  | 2017        | 81                 | US             | No*                          | 60       | 60       | Officer                 | 120                            | 0.51                                 | -                                            | Yes                        | Decrease                  | % change **                   |
| Headley et al. (2017)                                   | 2017        | 89                 | US             | Officer                      | 26       | 25       | Officer*time            | 102                            | 0.51                                 | -                                            | Yes                        | Null                      | t-test                        |
| Braga et al. (2018)                                     | 2018        | 180                | US             | Officer                      | 218      | 198      | Officer*time            | 832                            | 0.524                                | 1                                            | Yes                        | Decrease                  | DiD                           |
| Peterson et al. (2018)                                  | 2018        | 54                 | US             | Officer                      | 252      | 252      | Officer                 | 504                            | 0.5                                  | -                                            | Yes                        | Null                      | DiD                           |
| Wallace et al. (2018)                                   | 2018        | 107                | US             | Officer                      | 82       | 67       | Call-officer            | 228,220                        | 0.55                                 | 1+                                           | No                         | -                         | DiD                           |
| Yokum et al. (2019)                                     | 2019        | 91                 | US             | Officer                      | 1,189    | 1,035    | Officer                 | 1,922                          | 0.535                                | -                                            | Yes                        | Null                      | OLS                           |
| Koslitski et al. (2020)                                 | 2020        | 20                 | US             | -                            | -        | -        | Officer * Month         | -                              | -                                    | -                                            | Yes                        | Decrease                  | Time series analysis          |
| Braga et al. (2020)                                     | 2020        | 23                 | US             | Officer + District           | 140      | 141      | Officer                 | 562                            | 0.498                                | 1                                            | Yes                        | Decrease                  | DiD                           |
| <i>Panel C: Spatially explicit designs (12 studies)</i> |             |                    |                |                              |          |          |                         |                                |                                      |                                              |                            |                           |                               |
| Katz et al. (2014)                                      | 2014        | 217                | US             | Area                         | 1        | 1        | Area                    | 2                              | 0.5                                  | -                                            | No                         | -                         | t-test                        |
| Morrow et al. (2016)                                    | 2016        | 149                | US             | Area                         | 1        | 1        | IPV events              | 2,063                          | -                                    | -                                            | No                         | -                         | t-test**                      |
| Ariel (2016a)                                           | 2016        | 109                | US             | District                     | 1        | 5        | Street segment          | 17,726                         | 0.167                                | 2                                            | No                         | -                         | Unclear                       |
| Ariel (2016b)                                           | 2016        | 129                | US             | District                     | 1        | 5        | Call                    | 924,457                        | 0.167                                | -                                            | Yes                        | Null                      | Odds-Ratio                    |
| Hedberg et al. (2017)                                   | 2017        | 209                | US             | Precinct                     | 1        | 1        | Incident                | 44,380                         | 0.499                                | 1+                                           | No                         | -                         | GLM                           |
| Mitchell et al. (2018)                                  | 2018        | 19                 | UY             | Region                       | 5        | 14       | Region                  | 38                             | 0.263                                | -                                            | No                         | -                         | Time series analysis          |
| Bennett et al. (2019)                                   | 2019        | -                  | US             | Squad areas                  | 1        | 1        | Squad * Week            | Unclear                        | 0.5                                  | -                                            | Yes                        | Null                      | Diff. in trends test          |
| Kim (2024)                                              | 2021        | 11                 | US             | Agencies                     | 25       | 75       | Agency * Month          | 100                            | -                                    | -                                            | Yes                        | Decrease                  | TWFE                          |
| Miller and Chillar (2021)                               | 2021        | 10                 | US             | Agencies                     | 1,346    | 1,030    | Agency * Year           | 10,448                         | -                                    | -                                            | No                         | -                         | DiD                           |
| Bollman (2021)                                          | 2021        | -                  | US             | Courts                       | 70       | 33       | Court * Quarter         | 4,141                          | -                                    | -                                            | No                         | -                         | TWFE                          |
| Çubukcu et al. (2021)                                   | 2021        | 9                  | US             | District                     | -        | -        | Complaints              | 2,117                          | -                                    | -                                            | No                         | -                         | TWFE                          |
| Ferrazares (2023)                                       | 2023        | 1                  | US             | District                     | -        | -        | District * Day          | 46,011                         | -                                    | -                                            | Yes                        | Decrease                  | TWFE                          |

*Notes:* This table provides a non-exhaustive overview of some of the existing empirical literature on BWC. The overview does not claim to be comprehensive but has aimed to include all empirical studies evaluating BWCs across a broad range of fields from criminology to economics. In column (4), we indicate with \* if the evaluation includes volunteer officers. The table focuses on the respective randomization design, the outcome measurement approach, empirical strategy employed and whether effects on use of force (UoF) have been identified. Empirical strategies that do not perform statistical inference have \*\*. Not in all cases was it possible to infer all required input and only two papers have replication data available.

**Table A10:** The effects of BWC adoption on the number of events

|                                 | Number of<br>Events | Number of<br>Telephone<br>Initiated<br>Events | Number of<br>Officer<br>Initiated<br>Events |
|---------------------------------|---------------------|-----------------------------------------------|---------------------------------------------|
|                                 | (1)                 | (2)                                           | (3)                                         |
| Treated Precinct x Post         | 0.705<br>(0.762)    | 1.033<br>(0.726)                              | -0.226<br>(0.180)                           |
| Mean Dep. Var. Control Precinct | 19.163              | 17.145                                        | 2.000                                       |
| N                               | 7372                | 7372                                          | 7372                                        |

*Notes:* This table shows the effects of adopting BWCs on police activity at the precinct level. The data is at the precinct-day level, and there are 38 total precincts, of which five adopted BWC. The specifications include precinct, week and day of the week fixed effects. The regression uses number of officers in each precinct as weights. Standard errors are clustered at the precinct-by-week level. \*\*\* p<0.01; \*\* p<0.05; \* p<0.1.

**Table A11:** Registered Hypotheses in the Nov 2018 PAP

|           | <b>Main Hypothesis</b>                                                    | <b>Sub-hypothesis</b>                                                                                                                                                            |
|-----------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>H1</b> | BWC reduces use of force incidents                                        | by a) officers wearing a camera; b) officers in the same patrol group as those wearing a camera; c) officers attending an event where at least one officer was wearing a camera. |
| <b>H2</b> | BWC reduce citizens complaints against officers                           | by a) officers wearing a camera; b) officers in the same patrol group as those wearing a camera; c) officers attending an event where at least one officer was wearing a camera. |
| <b>H3</b> | BWC reduce use of force incidents by police officers that had in the past | a) worn a camera; b) patrolled with officer that had worn a camera; c) attended an event where one officer was wearing a camera.                                                 |
| <b>H4</b> | BWC reduce dispatch time                                                  | a) wearing a camera; b) patrolling with an officer wearing a camera.                                                                                                             |

**Table A12: Outcome Breakdown**

|                               | Use of Force         | Physical Force       | Non-Lethal Force  | Lethal Force       |
|-------------------------------|----------------------|----------------------|-------------------|--------------------|
|                               | (1)                  | (2)                  | (3)               | (4)                |
| Treated Event                 | -0.425***<br>(0.157) | -0.325***<br>(0.122) | -0.133<br>(0.108) | -0.132*<br>(0.067) |
| Mean Dep. Var. Control Events | 0.694                | 0.457                | 0.311             | 0.110              |
| <i>N</i>                      | 13274                | 13274                | 13274             | 13274              |

*Notes:* Table presents results on the impact of a body worn camera being present at a police event over measures of use of force, which are broken by category. The dependent variables are (i) “Use of Force ” is an indicator if there was any deployment of physical, non-lethal (mechanical) or lethal force by the police, not considering use of handcuff or arrest; (ii) (ii) “Physical Force” is an indicator if there was use of officers physical restraint on an individual, possibly involving using or hands and/or feet; (iii) “Non-Lethal Force” or mechanical force, which includes pepper sprays, tasers, batons and/or non-lethal shooting; (iv) “Lethal Force ” which includes the the deployment of firearms. All dependent variables are multiplied by 100. Specifications include police precinct-by-week, day of the week, number of officers and stratification bins fixed effects. Shifts without camera are excluded from the regression. Standard errors are clustered at the precinct-by-week level. \*\*\* p<0.01; \*\* p<0.05; \* p<0.1.

**Table A13: MVPF of BWC implementation**

|                                              | Incidence per 100k-year | Benefit of aversion | Estimates Table 2    |                              |                | Estimate Table A12   |                              |                |
|----------------------------------------------|-------------------------|---------------------|----------------------|------------------------------|----------------|----------------------|------------------------------|----------------|
|                                              |                         |                     | Estimated effect (%) | Averted events per 100k-year | Total benefits | Estimated effect (%) | Averted events per 100k-year | Total benefits |
|                                              | (1)                     | (2)                 | (3)                  | (4)                          | (5)            | (6)                  | (7)                          | (8)            |
| <b>A. Police Lethality</b>                   |                         |                     |                      |                              |                |                      |                              |                |
| Lethal use of force                          | 1.685                   | 816,743.99          | -0.612               | 1.032                        | 842,772.12     | -1.200               | 1.685                        | 1,376,197.30   |
| <b>B. Non-lethal Use of Force</b>            |                         |                     |                      |                              |                |                      |                              |                |
| Lethal use of force without fatality         | 12.13                   | 5,883.33            | -0.612               | 7.428                        | 43,703.91      | -1.200               | 12.13                        | 71,365.91      |
| Non-lethal                                   | 28.80                   | 5,883.33            | -0.612               | 17.637                       | 103,763.55     | -0.428               | 12.316                       | 72,461.38      |
| Physical                                     | 48.08                   | 1,315.10            | -0.612               | 29.444                       | 38,721.44      | -0.711               | 34.193                       | 44,966.51      |
| <b>C. Other Benefits Beyond Use of Force</b> |                         |                     |                      |                              |                |                      |                              |                |
| Arrest and handcuffs                         | 559.34                  | 200.73              | -0.059               | 32.981                       | 6,620.15       | -0.059               | 32.981                       | 6,620.15       |
| <b>D. Total benefits, costs, and MVPF</b>    |                         |                     |                      |                              |                |                      |                              |                |
| Total benefit                                |                         |                     |                      |                              | 1,035,581.17   |                      |                              | 1,571,611.25   |
| Total cost                                   |                         |                     |                      |                              | 111,815.84     |                      |                              | 111,815.84     |
| MVPF                                         |                         |                     |                      |                              | 9.26           |                      |                              | 14.06          |

*Notes:* This table the MVPF of implementing BWC, considering the estimates of our experiment. Column (1) shows the incidence of the events per 100k inhabitants and year, calculated with the pre-experimental data. Column (2) presents the monetary benefit of averting each event, using WTP estimates from [Cohen and Piquero \(2009\)](#) rescaled for the Brazilian context. Column (3) shows the estimates of BWC percentage change on outcomes from Table 2, and column (6) uses the estimates from Table A12 except for the outcome Arrests and handcuffs in Panel C, which is obtained always from our main regression results in both columns. Columns (4) and (7) show the number of events averted per 100,000 inhabitants-year. Columns (5) and (8) show the total benefits for society of each entry. Panel A considers the reduction of police lethality considering the likelihood that a death by the police could occur in events with lethal use of force. The likelihood that lethal use of force translates into police killings uses data from [Fórum Brasileiro de Segurança Pública \(2018\)](#) on deaths by the police for the city of Florianópolis in 2018. The value of a statistical life in Brazil is 816,743.99 in 2018 US dollars, the median value in the literature for Brazil VSL ([Corbi et al., 2006](#); [Cerqueira, 2014](#); [Rocha and Soares, 2015](#); [Viscusi and Masterman, 2017](#); [Cerqueira et al., 2020](#); [Pereira et al., 2020](#)). Panel B shows the benefits of averting non-lethal use of force cases, such lethal use of force without fatality (officer shoots but doesn't kill), non-lethal and physical uses of force. Panel C quantifies the benefits of BWC in decreasing the number of handcuffs and arrests from the victim perspective. Panel D shows the total benefit, total cost, and resulting MVPF from the intervention. The yearly total cost per 100k inhabitants of BWC is 111,815.84 USD which includes purchase costs, training, charging/footage downloading devices, as well as personnel cost required to operate the cameras. The MVPF is the ratio between the total benefit calculated with the WTP and the total cost.

**Table A14:** Total costs of BWC implementation

| <b>A. Costs of the experiment</b>                    |                  |              |              |
|------------------------------------------------------|------------------|--------------|--------------|
| <b>Item</b>                                          | <b>Unit cost</b> | <b>Units</b> | <b>Cost</b>  |
| Camera                                               | 341.62           | 73           | 24,938.26    |
| Docking stations                                     | 3,030.50         | 7            | 21,213.50    |
| Senior officer yearly wage                           | 127,881.72       | 1            | 127,881.72   |
| Junior officer yearly wage                           | 90,426.40        | 1            | 90,426.40    |
| Officer training and camera handling                 | 90,880.71        | 1.5          | 136,321.06   |
| <b>B. Total costs</b>                                |                  |              |              |
| Per year                                             |                  |              | 400,780.93   |
| Per treated officer-year                             |                  |              | 2,671.87     |
| Scaled to 100% treatment (per year)                  |                  |              | 1,210,358.42 |
| Scaled to 100% treatment (per 100k inhabitants-year) |                  |              | 111,816      |

*Notes:* Panel A summarizes the costs of our experiment per item, including the cameras, the docking stations for charging, personnel and training costs. We summarize these costs in Panel B, considering the total cost for one year of experiment, this cost per treated officer-year, dividing the total cost by 150. We scale the total costs to have all officers in the precincts wearing a BWC. The last considered value considers the total costs scaled to 100% per 100k inhabitants-year, dividing the previous value by the population covered by the precincts.